# Structural Gaps in Urban Harm Reduction Infrastructure: A Block-Level Access and Policy Analysis of Dallas County, Texas

Ali I. Ilyas[1,2] and Sameer Ilyas[1]

[1]Department of Research, SaveSpots National Harm Reduction Program, Chicago, Illinois

[2]Department of Health, Oak Cliff Empowerment Center, Dallas, Texas

SaveSpots Research Program 2026 Cohort — City-Level Harm Reduction Research

July 2026

## Author Note

Ali I. Ilyas is affiliated with the SaveSpots National Harm Reduction Program (Chicago, Illinois), Department of Research, and with the Oak Cliff Empowerment Center, Department of Health, Dallas, Texas.

Sameer Ilyas is affiliated with the SaveSpots National Harm Reduction Program (Chicago, Illinois), Department of Research.

**Author contributions.** A.I.I. conducted the policy analysis, the comparative case analysis, and the gap identification, and drafted the manuscript. S.I. designed and implemented the geospatial data pipeline, the measured block-level access analysis, the composite vulnerability index, and the placement model, and produced the figures. Both authors interpreted the findings, revised the manuscript critically for important intellectual content, approved the final version, and agree to be accountable for all aspects of the work.

**Ethics.** This study analyzed only publicly available, de-identified, aggregate secondary data. It does not constitute human subjects research and did not require institutional review board approval. No individual-level records were obtained, held, or analyzed at any stage.

**Funding.** This research received no external funding.

**Competing interests.** The authors declare no competing interests. Both authors are affiliated with SaveSpots, an organization engaged in community naloxone distribution, and that affiliation is disclosed here in full; the analysis reports findings without recommending action by any specific organization.

**Data availability.** All analytic outputs, including block-level access distances, composite index scores, tier boundary definitions, candidate rankings, and figure exports, are openly available at https://github.com/Ilyas1434/dallas-research-savespots and permanently archived at

https://doi.org/10.5281/zenodo.21537399. All primary inputs are public datasets identified with accession identifiers and retrieval dates in Section 2.

**Use of artificial intelligence.** AI-assisted tooling was used for analytic pipeline construction, figure generation, and drafting support. All statistics derive from the cited primary sources; the authors verified all reported values against those sources and take full responsibility for the content.

**Acknowledgments.** The authors thank Musab Abdali, Esq., Board Chairman, Oak Cliff Empowerment Center, for institutional support and review.

**Correspondence.** Sameer Ilyas, SaveSpots National Harm Reduction Program. Email: research@savespots.org

## Abstract

**Background:** Dallas County, Texas recorded a 77.1% increase in drug overdose deaths between 2016 and 2024, with fentanyl-related deaths rising 1,550% over the same period. The county operates within one of the most restrictive harm reduction policy environments in the United States, lacking a comprehensive Good Samaritan law, authorized syringe services programs, legal fentanyl test strips, and Medicaid expansion. Texas mortality confidentiality rules suppress overdose death counts at fine geographic resolution, which constrains conventional hotspot analysis.

**Methods:** This analysis inverts the standard approach by measuring access rather than modeling risk. Straight-line distances from all 25,682 populated 2020 census blocks to all 38 fixed naloxone access points were computed in EPSG:32138 with no interpolation or smoothing. A five-layer, equal-weighted composite vulnerability index (model-based overdose mortality, poverty, uninsurance, measured supply distance, inverted transit connectivity) was constructed at census tract level with Jenks natural-break tiering and two-variant sensitivity analysis. Candidate storefronts were ranked by need multiplied by measured reach among currently unserved residents. Houston and Philadelphia served as structurally analogous and best-practice comparison cases.

**Results:** Of 2,613,539 county residents, 2,196,211 (84.03%) live more than one mile from any fixed naloxone access point, 1,506,855 (57.66%) beyond two miles, and 902,611 are stranded beyond both one mile of supply and one-half mile of transit. Exactly one fixed 24-hour access point exists countywide, and 86.25% of residents live more than five miles from it. Of 643 ranked tracts, 170 fall in the highest vulnerability tier, driven more by transit inaccessibility (38.2% of composite) than by modeled mortality (15.2%). Twenty-five optimally selected,

non-overlapping storefronts would newly place supply within a half-mile walk of 231,806 currently unserved residents.

**Conclusions:** Dallas County's naloxone access deficit is measurable, large, and concentrated in identifiable corridors. Because mortality is suppressed at fine geography while access is exactly measurable, coverage of unserved residents is the more defensible optimization target. Embedded distribution in everyday commercial settings is permissible under existing Texas standing order law.

*Keywords:* harm reduction; naloxone access; opioid overdose; spatial accessibility; data suppression; Texas health policy; geographic health disparities; overdose prevention

# Section 1: Introduction

## 1.1 Background and Significance

The United States has experienced four distinct waves of opioid-related overdose mortality since the late 1990s, beginning with widespread prescribing of commercial opioids, followed by a shift to heroin, the emergence of illicitly manufactured fentanyl as the dominant driver of overdose deaths, and most recently a rise in polydrug fatalities involving fentanyl combined with stimulants such as cocaine and methamphetamine (Federal Reserve Bank of Dallas, 2025). National provisional drug overdose deaths peaked at 110,037 in 2023 before declining approximately 27% to 80,391 in 2024, the largest single-year reduction recorded in the modern era of the overdose crisis (Centers for Disease Control and Prevention, National Center for Health Statistics [CDC NCHS], 2026a). This decline has been attributed to expanded naloxone access, increased availability of medications for opioid use disorder, harm reduction infrastructure expansion in states with permissive policy environments, and possible changes in illicit fentanyl supply characteristics (Brookings Institution, 2025).

The decline has not been uniform. Jurisdictions operating within restrictive harm reduction policy environments, specifically those without comprehensive Good Samaritan laws, authorized syringe services programs, legal fentanyl test strips, or Medicaid expansion, have experienced substantially less improvement than jurisdictions implementing the full complement of evidence-based tools. Texas maintains some of the most restrictive harm reduction policies in the country and continues to operate without the policy infrastructure associated with the largest mortality reductions elsewhere (Tough, 2025).

The evidence base for harm reduction as a public health strategy is extensive and consistent. Community naloxone distribution is associated with measurable reductions in

overdose mortality (Fischer et al., 2025). Syringe services programs reduce HIV and hepatitis C transmission without increasing drug use in the communities they serve (Centers for Disease Control and Prevention [CDC], 2024). Good Samaritan laws are associated with reductions in overdose deaths of up to 15% and with increased likelihood of emergency calls during overdose events (U.S. Government Accountability Office [GAO], 2021). Medication-assisted treatment reduces all-cause mortality, overdose risk, and criminal justice involvement among people with opioid use disorder (Kaiser Family Foundation [KFF], 2025).

### 1.2 Gap in the Literature

Peer-reviewed literature on urban harm reduction infrastructure has predominantly examined cities operating within permissive state policy environments, including San Francisco, New York, and Philadelphia, where the range of available interventions is comparatively broad (Claborn et al., 2023; Walley et al., 2013). Substantially less attention has been directed toward counties operating under highly restrictive state frameworks, where state-level prohibition, limited local public health capacity, and community-level overdose risk interact to produce a distinct set of structural challenges.

A second and less examined gap is methodological. Spatial prioritization in harm reduction typically proceeds by modeling risk, most commonly by mapping overdose mortality and identifying areas of elevated rate. In states where mortality confidentiality rules suppress death counts at fine geography, that approach either fails or proceeds on smoothed model-based estimates whose uncertainty is rarely surfaced. Analyses that instead treat spatial access as the primary measured quantity, and reserve modeled mortality for a secondary role, remain uncommon in this literature. Where suppression is severe, the distinction between what has been counted and what has been estimated becomes consequential for resource allocation.

### 1.3 Research Objectives and Contribution

This analysis addresses five objectives: (1) characterizing Dallas County's overdose burden relative to state and national trends using both provisional and finalized surveillance products; (2) evaluating the policy environment governing harm reduction access in Texas; (3) measuring, rather than estimating, the population-weighted distance between county residents and existing naloxone supply; (4) constructing a transparent multi-layer vulnerability index in which modeled mortality is one input among five, with disclosed weights and reported sensitivity; and (5) identifying candidate sites at which additional supply would reach the largest number of currently unserved residents.

The analysis contributes, to the authors' knowledge, the first block-level measured naloxone access assessment for Dallas County, and a reporting convention that classifies every quantity as measured, modeled, or published so that readers can distinguish counted facts from statistical estimates. Where policy constraints limit the range of available interventions, the analysis documents both the nature of those constraints and the evidence base supporting their modification.

## Section 2: Methodology

### 2.1 Study Design and Evidence Classification

This analysis employs a secondary data synthesis design integrating three components: review of gray and peer-reviewed literature, secondary analysis of government surveillance and administrative data, and geospatial measurement of service access.

Every quantitative claim reported in this paper is classified into one of three categories, and the classification is stated wherever ambiguity could arise:

*Measured.* A direct count of people or a computed geometric distance, involving no modeling, interpolation, or smoothing. This category comprises 2020 decennial census block populations, Euclidean distances computed in EPSG:32138, candidate reach counts, and supply point geometry.

*Modeled.* Model-based rates published by CDC NCHS, survey estimates from the American Community Survey, and composite scores or tiers derived from them.

*Published.* A figure transcribed from a cited government report, principally the Dallas County Health and Human Services (DCHHS) Overdose Data to Action: 2024 Annual Surveillance Report (DCHHS, 2024), with the corresponding figure or page reference.

Where a quantity could not be produced from open data, it is reported as not computable and attributed to its external source rather than estimated. No original primary data were collected by the authors. Dallas County-specific primary qualitative findings were drawn from the Baumgartner and Ledat (2024) Community Needs Assessment, which employed stakeholder focus groups and semi-structured interviews with people with lived and living experience of drug use; those findings are incorporated in Sections 5 and 7. All statistics were triangulated against a minimum of two independent sources where two exist; where sources conflicted, the more conservative estimate was used and the discrepancy is noted.

**Ethics statement.** This study analyzed only publicly available, de-identified, aggregate secondary data. It does not constitute human subjects research and did not require institutional review board approval.

### 2.2 Mortality Data Sources

Two mortality products are used and are not interchangeable. CDC NCHS Vital Statistics Rapid Release (VSRR) provisional counts (Socrata dataset gb4e-yj24; data as of July 5, 2026)

provide county-level 12-month-ending totals that revise upward as toxicology and death certificate processing complete. DCHHS Overdose Data to Action (2024) provides calendar-year counts derived from Texas Department of State Health Services Vital Statistics as of August 14, 2025, with rates from CDC WONDER as of April 2, 2025, and emergency department figures from ESSENCE.

These products disagree by construction. Calendar year 2024 deaths are reported as 558 by DCHHS, while the VSRR 12-month period ending December 2024 records 566. Neither figure is incorrect; they are different instruments measuring overlapping but non-identical windows. Throughout this paper, each mortality figure is attributed to its source and window, and the two series are never combined.

Sub-county modeled rates are drawn from the CDC NCHS model-based drug overdose death rate product (Socrata dataset 4day-mt2f; 12-month period January 2025 through December 2025; data as of July 22, 2026).

### 2.3 The Suppression Regime and Its Consequences for Method

Texas mortality confidentiality rules materially constrain fine-grained overdose analysis, and that constraint determines this study's design.

First, count suppression is near-universal at tract level. The CDC NCHS product publishes model-based rates for all 645 Dallas County tracts but withholds underlying death counts wherever the numerator falls between 1 and 9. In Dallas County, 592 of 645 tracts (91.8%) carry suppressed counts. These are not zero-death tracts; they are low-numerator tracts whose true burden is statistically invisible.

Second, the published geography is coarser than the tract. CDC does not model at raw tract level. It publishes at 150 native aggregation units for Dallas County, each a union of

approximately 2 to 10 tracts (mean 4.3), and each carrying a published 95% confidence interval. Of those 150 units, 138 (92%) have suppressed underlying counts. Distributing one unit's modeled rate across its constituent tracts, as is common practice, visually smears variance that the source data does not resolve.

Third, no ZIP-level product exists to suppress. Prior analyses of Dallas County have described a specific number of ZIP codes as suppressed by the state. That framing is superseded: no current public Texas Department of State Health Services ZIP-level overdose mortality product exists at all. Confidentiality suppression now manifests as tract count suppression in the federal product rather than as ZIP-level withholding in a state product.

Given these constraints, modeling risk at fine geography would require inference layered on inference. This analysis therefore treats spatial access as the primary measured quantity. Residents can be counted exactly at block level, and the distance from each block to each supply point is a matter of geometry rather than estimation. Modeled mortality enters the analysis only as one of five equally weighted inputs to a tract-level composite, and its limitations are carried forward explicitly.

### 2.4 Measured Access Analysis

The measured access analysis uses 2020 Census P.L. 94-171 block populations (variable P1_001N) joined to TIGER/Line 2024 tabulation blocks (U.S. Census Bureau, 2021, 2024). Dallas County contains 38,180 blocks, of which 25,682 are populated, with a total 2020 decennial population of 2,613,539. Block locations are Census Bureau internal points (INTPTLAT20, INTPTLON20).

Naloxone supply locations were assembled from the Texas Department of State Health Services NarcanSites feed (April 2026), a local verified inventory, and the SAMHSA treatment

facility locator (Substance Abuse and Mental Health Services Administration [SAMHSA], 2026), yielding 39 sites. Of these, 38 have usable fixed coordinates and are included; one 24-hour mobile harm reduction unit operates without fixed coordinates and is excluded, since a moving unit cannot be represented as a fixed catchment. The 38 fixed sites comprise 13 opioid treatment programs or methadone clinics, 10 storefront dispensers, 7 nightlife venues, 4 recovery service locations, 3 harm reduction access points, and 1 health or county clinic.

All distances are straight-line (Euclidean), computed in EPSG:32138 (NAD83 / Texas North Central, metre units) via a k-dimensional tree nearest-neighbor search. A resident is classified as beyond a given threshold if the internal point of their block lies farther than that distance from the nearest relevant supply site. Thresholds are 1 mile (1,609.344 m), 2 miles (3,218.688 m), and 5 miles (8,046.72 m). Transit proximity uses 805 m (one-half mile) from any of the 6,974 Dallas Area Rapid Transit (DART) stops in the July 2026 General Transit Feed Specification release.

A block is classified as *stranded* when three measured conditions hold simultaneously: population greater than zero, nearest supply point beyond 1,609.344 m, and nearest DART stop beyond 805 m. Every term in that definition is a count or a distance.

Straight-line distance understates real travel burden in a county with major highway severance, the Trinity River floodplain, and the Great Trinity Forest separating southeastern communities from central infrastructure. Reported deserts are therefore conservative lower bounds on hardship. A second and more consequential conservatism is that the 38 supply points include facility types whose walk-in public access is materially limited: opioid treatment programs dispense methadone under federal certification rather than distributing naloxone to the

public, and nightlife venue dispensers operate only during business hours. Crediting these locations as supply makes the measured desert smaller than the functional desert.

### 2.5 Composite Vulnerability Index

The composite index operates at census tract level (TIGER/Line 2024, 2020 boundaries), the finest geography at which all inputs are natively published. Of 645 tracts, 643 are ranked; two zero-population tracts are excluded.

Five layers are combined (Table 1):

**Table 1**

*Composite Vulnerability Index: Layer Sources, Vintage, and Classification*

| Layer | Source | Vintage | Class |
|---|---|---|---|
| Overdose mortality rate per 100,000 | CDC NCHS 4day-mt2f | Jan–Dec 2025; as of 2026-07-22 | Modeled |
| Poverty rate | ACS S1701_C03_001E | ACS 2020–2024 5-year | Modeled (survey) |
| Uninsured rate | ACS S2701_C05_001E | ACS 2020–2024 5-year | Modeled (survey) |
| Distance to nearest naloxone supply | This analysis | 2026-07 | Measured |
| DART stops within 0.5 mi (inverted) | DART GTFS | 2026-07 | Measured |

Each layer was normalized to a 0 to 1 scale by min-max normalization. Layers were combined with equal weights of 0.20 each, following the methodological convention of the CDC/ATSDR Social Vulnerability Index technical documentation (Agency for Toxic Substances and Disease Registry [ATSDR], 2022) and disclosed here as an analyst choice rather than an

empirically derived weighting. The composite for each tract is the weighted mean over non-null layers; tracts with partial coverage were rescaled and flagged.

Two candidate layers were deliberately excluded. Non-fatal overdose emergency department visits and point-in-time homelessness counts are published only at county or continuum-of-care level, and allocating them to tracts would constitute ecological fabrication. Their exclusion is a documented limitation rather than an oversight.

Priority tiers were assigned by Jenks natural-break classification into three classes, a method used in public health geographic prioritization to identify discontinuities in the observed distribution rather than imposing arbitrary quantile boundaries (Jenks, 1967). Resulting cut points are 0.338 (Tier 3 to Tier 2) and 0.434 (Tier 2 to Tier 1).

**Sensitivity analysis.** Two variants re-ran the full pipeline with renormalized weights and recomputed Jenks breaks, compared tract by tract against the primary specification. Doubling the mortality weight left 86.94% of tracts (559 of 643) in the same tier, with 84 flips. Doubling the distance weight left 78.54% (505 of 643) in the same tier, with 138 flips, or 21.46%. The distance-weight variant exceeds a 15% flip threshold and is flagged accordingly: the Tier 1 set is one defensible configuration, not a robust one. Under both variants, every flip occurred between adjacent tiers, and no tract moved between Tier 1 and Tier 3 under either specification, indicating that the coarse high-versus-low separation is stable even where exact tier membership is not.

### 2.6 Placement Model

Candidate host locations were assembled from Dallas County business records, Texas Alcoholic Beverage Commission off-premise permits (Texas Alcoholic Beverage Commission [TABC], 2026), OpenStreetMap points of interest, and Dallas Public Library branch locations,

yielding 5,538 candidates. Of these, 5,428 were ranked; 110 were excluded for falling outside county boundaries or lying in zero-population tracts.

*Reach* (measured) is the sum of 2020 block population over blocks that are both currently unserved (beyond 1,609.344 m from existing supply) and within 805 m of the candidate, that is, the number of currently unserved residents within a half-mile walk of that storefront. Reach is right-skewed: mean 2,994.3, median 2,774.5, maximum 17,437, with 773 candidates having reach of zero.

*Need* was operationalized two ways. The composite variant uses the five-layer index described above, which contains modeled mortality. The grounded variant is the equal-weighted mean of normalized poverty rate, uninsured rate, Social Vulnerability Index, and measured share of tract population beyond one mile from supply, and contains no mortality input at all. Each need variant was normalized and multiplied by normalized reach; raw reach is retained alongside so that no information is lost.

The two rankings agree closely (Spearman rho = 0.9659; 45 of the top 50 and 91 of the top 100 shared). This agreement is itself a finding: the placement recommendation does not materially depend on the suppressed, modeled mortality layer.

A separate rank-agnostic greedy coverage analysis was also run, iteratively selecting the site covering the largest number of not-yet-covered unserved residents within 805 m, removing those blocks, and repeating. This produces de-overlapped marginal coverage and is reported separately from the ranked list because the two orderings differ.

A limitation applies to candidate categorization. Category labels derive from license class and point-of-interest tags rather than from field verification. TABC off-premise permit classes in particular do not correspond to colloquial storefront type, so a restaurant holding an off-premise

permit may appear under a beverage category. Categories should be read as license-derived descriptors, and final siting requires ground-level verification.

### 2.7 Comparative Analysis

Two purposively selected comparison cases were used. Houston, Texas was selected as a structurally analogous case on the basis of demographic and structural parallels within the same state regulatory environment, allowing policy restrictiveness rather than demographic or economic conditions to function as the primary differentiating variable. Philadelphia, Pennsylvania was selected as a best-practice case on the basis of comparable urban poverty concentration, majority-minority demographic composition, and documented severe overdose burden, combined with the broadest implementation of evidence-based harm reduction tools among major U.S. cities. These cases were not randomly assigned and do not constitute controlled conditions; confounders including political leadership, private foundation investment, and drug supply variation across periods cannot be fully accounted for. The comparison is presented as structured qualitative evidence rather than causal inference.

### 2.8 Supporting Context Data

Poverty, uninsurance, and ethnicity estimates are drawn from the American Community Survey 2020–2024 5-year release (U.S. Census Bureau, 2026). Social vulnerability is drawn from the CDC/ATSDR Social Vulnerability Index 2022 (ATSDR, 2022), which ranks Dallas County 71st of 254 Texas counties. Homelessness figures are drawn from the Housing Forward North Texas 2026 point-in-time count (Housing Forward North Texas, 2026). Built-environment indicators use Dallas 311 service requests from the preceding 24 months; a companion municipal code violations archive was found to terminate in July 2018 and was therefore not used for current-condition inference.

### 2.9 Limitations

Several limitations apply and are carried into the interpretation of every finding.

Suppression is extensive. In 592 of 645 tracts (91.8%) and 138 of 150 native reporting units (92%), underlying death counts are suppressed, including 149 of the 170 Tier 1 tracts. The highest-priority designations therefore rest on modeled rather than observed mortality.

Provisional counts revise upward. VSRR points for the most recent approximately six months are depressed by a reporting lag of roughly seven months. Second-half 2025 counts should be read as floors rather than settled values. DCHHS similarly labels its 2024 figure provisional pending certification.

Distances are Euclidean. Actual walking and driving distances are longer, so reported deserts understate hardship.

One 24-hour mobile unit is unmeasurable. The reported 24-hour desert is measured against a single fixed point; were the mobile unit's routes geocoded, that desert could shrink.

Certain metrics are not computable from open data. Concordance between decedent residence ZIP and incident ZIP, and the proportion of overdose death locations within walking distance of supply, both require case-level medical examiner records available only by request. These are reported as gaps rather than estimated.

Modeled tract rates are smeared. One modeled rate is applied across an average of 4.3 tracts, so tract-level variance in Figure 2 is an artifact of disaggregation rather than an observed pattern.

Composite tiering is sensitive to distance weighting, as reported in Section 2.5.

Candidate categories are license-derived and unverified in the field.

No original primary data were collected by the authors. The analysis incorporates Dallas County-specific primary qualitative findings from Baumgartner and Ledat (2024), which partially addresses the community voice gap but does not substitute for participatory or ethnographic research directly engaging people who use drugs in intervention design. Future research should integrate participatory methods with community members in the county's highest-priority zones.

## Section 3: Dallas County Profile

### 3.1 Demographics

Dallas County is the second most populous county in Texas. The 2020 decennial census records an exact population of 2,613,539, while the ACS 2020–2024 5-year estimate places it at 2,604,053 (U.S. Census Bureau, 2021, 2026). The decennial count is used throughout the access analysis because it is an exact enumeration at block level; the ACS estimate is used where survey-derived characteristics are required. The city of Dallas holds approximately half the county's residents.

Hispanic or Latino residents constitute 41.3% of the county population (1,081,496 of 2,621,179; ACS table B03003, retrieved July 23, 2026; table-specific universes in the ACS differ slightly from the countywide estimate reported above). DCHHS independently reports that approximately 40% of Dallas County households speak a language other than English, with roughly one in five of those households having limited English proficiency, the highest such proportion in the region (DCHHS, 2024, p. 3). This linguistic profile has direct consequences for harm reduction service design, examined in Section 5.4.

Approximately 21% of county residents lack health insurance (ACS S2701_C05_001E, 2020–2024 5-year; U.S. Census Bureau, 2026), among the highest uninsured rates of any large

urban county in the United States. Because buprenorphine and methadone treatment generally require clinical entry points, this rate functions as a direct constraint on treatment access.

The Housing Forward North Texas 2026 point-in-time count recorded 3,513 individuals experiencing homelessness across Dallas and Collin counties on a single night, with 82% located in Dallas County (Housing Forward North Texas, 2026). Because this count is published only at continuum-of-care level, it is reported here as context and was excluded from the tract-level composite (Section 2.5).

### 3.2 Overdose Mortality

Two series describe county mortality and are reported separately.

**Provisional county series (VSRR).** Across 72 monthly 12-month-ending observations from January 2020 through December 2025, deaths ranged from 352 (period ending January 2020) to 668 (period ending February 2024). The most recent value is 582 for the 12 months ending December 2025, representing a 12.9% decline from the peak and a 2.8% increase over the same period ending December 2024 (566). Crude rates computed against the ACS county population are 17.36 per 100,000 in 2020, 21.66 in 2021, 22.00 in 2022, 25.11 in 2023, 21.74 in 2024, and 22.35 in 2025. The computable peak change within this series is 2020 to 2023, a 44.7% increase in counts (452 to 654). All 2025 values are provisional and subject to upward revision.

**Finalized calendar-year series (DCHHS).** Annual death counts rose from 315 in 2016 to 558 in 2024, an increase of 77.1%, with a peak of 628 in 2023 and a subsequent decline of 11.1% (DCHHS, 2024, Fig. 1a, p. 4). Across the full 2016 to 2024 window, 4,028 residents died of drug overdose, of whom 2,455 (60.9%) resided within the City of Dallas (DCHHS, 2024, p. 9, Fig. 10). The crude mortality rate rose from 12.9 to 20.8 per 100,000 between 2018 and 2024, an

increase of 61%, compared with 49.5% statewide and 9.9% nationally over the same window (DCHHS, 2024, Fig. 1b, p. 4). The 61% figure and the VSRR-derived 44.7% figure describe different windows and different products and are not in conflict.

Substance-specific changes between 2016 and 2024 show fentanyl-related deaths increasing 1,550%, methamphetamine 319%, and cocaine 148%, while heroin declined 69% and treatment or prescription opioids declined 31% (DCHHS, 2024, Figs. 2–3, pp. 5–6). Opioid-stimulant combination deaths rose 260% and stimulant-only deaths 167% over the same period. Drug categories in these figures are not mutually exclusive.

Demographic patterns show the highest overall burden among males, White individuals, and adults aged 35 to 64 (DCHHS, 2024, p. 3). Deaths among males rose from 210 to 413 and among females from 105 to 145 between 2016 and 2024 (DCHHS, 2024, Fig. 4, p. 7). One shift carries particular relevance for service design: while fentanyl deaths across the full 2016 to 2024 window were highest among White males aged 25 to 34, in 2024 Hispanic males recorded the highest number of fentanyl-related deaths of any combined sex and race category (DCHHS, 2024, Figs. 11a–c, p. 10). Cocaine-related deaths are highest among Black males aged 55 to 64 (p. 12), and methamphetamine deaths highest among White males aged 55 to 64 while increasing among Hispanic males (p. 13). The three ZIP codes with the highest resident overdose death counts across the window are 75217, 75216, and 75215 (DCHHS, 2024, Fig. 9, p. 9).

### 3.3 Non-Fatal Overdose

All-drug overdose emergency department visits rose from 2,419 in 2018 to 3,863 in 2023 before declining to 3,663 in 2024, a net increase of 51.4% across the window and a 5.2% single-year decline (DCHHS, 2024, Fig. 14, p. 16). Opioid-related emergency department visits totaled 793 in 2024, down 5.7% from 841 in 2023 (DCHHS, 2024, Fig. 15a, p. 16). Among 2024

opioid-related visits, 69.2% involved individuals aged 18 to 44, with a male-to-female ratio of 2.1 to 1 (p. 17).

These emergency department figures and the mortality figures in Section 3.2 measure different phenomena and should not be read as contradictory when they move in different directions. Non-fatal events are also systematically undercaptured: nationally, non-fatal overdoses occur at a ratio of approximately 8.4 to every fatal overdose, and many never generate an emergency call or a hospital visit (Federal Reserve Bank of Dallas, 2025).

### 3.4 Treatment Access and Retention

Access to medication-assisted treatment in Dallas County is constrained by insurance gaps, provider capacity, and stigma. Nationally, only 10.38% of patients diagnosed with opioid use disorder received a medication prescription within 12 months of diagnosis (national benchmark; Liang et al., 2023). National retention in opioid treatment programs averaged 61% at 30 days, 41.5% at 90 days, and 27.5% at 180 days (national benchmark; Medscape, 2026). Because 21% of county residents are uninsured and methadone requires daily in-person clinic attendance, these national retention challenges are likely compounded locally by cost and transportation barriers. County-specific enrollment and retention figures are not published, and no local estimate is offered here.

### 3.5 Infectious Disease Burden

Infectious disease indicators provide an indirect measure of injection drug use prevalence and syringe access gaps. Dallas County continues to record new HIV cases at a steady rate, with an approximately 30% increase among males and 6.5% among females reported in 2021 (DCHHS, n.d.-b). A study at the Dallas County Jail found 16.4% of inmates positive for hepatitis C antibody, approximately nine times the general Texas population rate of 1.8% and consistent

with national correctional prevalence ranges of 12% to 35% (de la Flor et al., 2017). These figures carry particular weight in the absence of a county syringe services program, since injection drug use remains the primary transmission vector for both infections.

### 3.6 Criminal Justice Intersection

Texas maintains among the most punitive drug possession statutes in the country, and possession of small quantities can result in felony charges. Dallas County's DIVERT Court, a voluntary diversion program for first-time felony drug possession defendants, has demonstrated meaningful recidivism reduction: graduates showed re-arrest rates equal to 32% of the rate observed in the comparison group processed through traditional adjudication (Fomby, n.d.). More broadly, Texas drug court participants recorded re-arrest rates of approximately 39% compared with 59% for traditional processing (Fomby, n.d.). Diversion capacity remains limited relative to the volume of drug-related arrests.

### 3.7 Existing Harm Reduction Infrastructure

DCHHS is the primary public health anchor for harm reduction in the county, providing free naloxone to businesses, organizations, and community centers upon completion of a training program, with outreach directed at restaurants, bars, shelters, libraries, and recreation centers (DCHHS, n.d.-a). In October 2024, DCHHS launched an Overdose Prevention Hotline in partnership with Parkland Hospital and the North Texas Poison Center, operating from 7 a.m. to 11 p.m. daily. Call volume increased 75% between November 2024 and March 2025, indicating demand outpacing current capacity (CDC, 2025).

At state level, the Texas Targeted Opioid Response program implements prevention, treatment, and recovery strategies across community, clinical, and criminal justice settings (Texas Health and Human Services, n.d.), and the Texas Overdose Data to Action program

maintains a naloxone distribution map covering community health clinics, recovery support services, and regional distribution sites (Texas Department of State Health Services [DSHS], 2026).

Dallas County does not operate a syringe services program. Research consistently demonstrates that such programs decrease infectious disease transmission among people who inject drugs and increase treatment engagement (CDC, 2024). Given the county's HIV trajectory and the 16% hepatitis C prevalence recorded among its incarcerated population, the absence of an authorized program represents a consequential structural gap.

## Section 4: Policy Analysis

The Texas legal and regulatory environment defines the operational boundaries within which harm reduction must be strategized in Dallas County. Recent legislative activity signals growing political recognition of the crisis, but is counterbalanced by the absence of a comprehensive Good Samaritan law, statewide prohibition of syringe services programs, continued criminalization of fentanyl test strips, non-expansion of Medicaid, and severe drug possession penalties. Table 2 summarizes the status and analytical assessment of each policy area.

### 4.1 Naloxone Access Laws

Naloxone access in Texas operates through a physician-issued standing order framework. Texas Senate Bill 1462 authorized physicians to issue standing orders, enabling pharmacies and organizations to dispense naloxone without individual prescriptions (Texas Overdose Training Initiative [TXOTI], 2025). Any pharmacy or organization operating in Texas may request a free standing order through TXOTI, which covers distribution by any employee or volunteer of the requesting organization and provides legal protection for administration during a suspected

overdose (TXOTI, 2025). This framework is the legal foundation on which the placement analysis in Section 8.6 rests: a business hosting a naloxone supply point may lawfully possess and distribute naloxone under existing law, requiring no statutory change.

In practice, naloxone was accessible via standing order at 83.7% of audited Texas pharmacies (Hill et al., 2019). The 2023 federal approval of over-the-counter naloxone expanded access nationally, though a retail price point of roughly $40 to $50 has limited uptake, with most Texans obtaining naloxone through no-cost programs (Tough, 2025). More than 90% of the DCHHS harm reduction budget derives from federal funding sources now facing proposed reductions. That dependency presents a structural vulnerability capable of disrupting distribution infrastructure irrespective of the standing order framework's legal soundness (Tough, 2025).

**4.2 Good Samaritan Law**

Texas is one of only three states, with Kansas and Wyoming, lacking a comprehensive Good Samaritan law protecting individuals who call emergency services during a drug overdose (GAO, 2021). The 2021 Jessica Sosa Act provides a limited affirmative defense rather than immunity (Texas Health and Safety Code § 481.115(g)-(i), 2021). Its protections are narrowly bounded: the caller must be the first to call, must remain at the scene and cooperate with law enforcement, must not have invoked the defense within the prior 18 months, and must not have been in the process of arrest at the time of the call. Once used, the defense cannot be invoked again.

These constraints render the statute inadequate as a harm reduction instrument. GAO analysis found consistently lower opioid overdose death rates in states with robust Good Samaritan laws and increased likelihood of emergency calls where bystanders are aware of such protections (GAO, 2021). Law enforcement involvement is documented as a primary deterrent to

emergency calls, with fewer than half of witnessed overdoses resulting in emergency contact in states lacking comprehensive protections (GAO, 2021; Texas Criminal Justice Exchange, n.d.).

### 4.3 Fentanyl Test Strip Legality

Fentanyl test strips remain classified as drug paraphernalia under Texas law, carrying Class C misdemeanor penalties of up to $500 (KVUE News, 2025). The classification is consequential given the substance profile documented in Section 3.2, in which a substantial share of fentanyl-involved deaths occurred among individuals consuming cocaine, methamphetamine, or counterfeit pills rather than knowingly consuming opioids.

In April 2025 the Texas House of Representatives passed House Bill 1644 by a vote of 145 to 0, seeking to remove fentanyl and xylazine test strips from the paraphernalia classification (Texas Legislature, 2025). A comparable measure passed the House 143 to 2 in 2023 but failed to advance from the Senate Criminal Justice Committee (Texas Observer, 2025). House Bill 1644 followed the same trajectory, expiring in that committee at adjournment of the 89th legislative session. Continued criminalization of drug-checking technology maintains a structural barrier to a low-cost prevention intervention.

### 4.4 Syringe Access

Texas does not authorize statewide syringe services programs. Under the Texas Controlled Substances Act, hypodermic syringes are classified as drug paraphernalia when possessed with intent to inject a controlled substance, creating legal barriers to both distribution and possession outside licensed medical contexts (Texas Health and Safety Code § 481.002). The only county-level authorization in Texas applies to Bexar County, where a locally funded pilot was incorporated into state law in April 2025 (Filter Magazine, 2025). Pending legislation would

extend pilot authorization to six additional counties; Dallas County is not among them (Filter Magazine, 2025).

### 4.5 Medication-Assisted Treatment Environment

Methadone for opioid use disorder may be dispensed only through federally certified opioid treatment programs requiring daily in-person attendance, which presents a structural barrier for individuals without reliable transportation, stable housing, or flexible employment (NAMA Recovery, 2024). This constraint bears directly on the access analysis: the 13 opioid treatment programs counted among the 38 supply points in Section 5 do not function as public naloxone access points. Buprenorphine prescribing was liberalized by federal elimination of the DATA 2000 waiver requirement in 2023, expanding the potential prescriber pool substantially, though county-level capacity remains constrained by provider willingness and patient insurance status.

### 4.6 Drug Possession and Penalty Structure

Fentanyl holds a distinct classification under Penalty Group 1-B, with possession of trace amounts constituting a felony (Texas Health and Safety Code § 481.115). Penalties escalate from a state jail felony for less than one gram, to a third-degree felony carrying two to ten years for one to four grams, a second-degree felony carrying two to twenty years for four to 200 grams, a first-degree felony carrying up to 99 years for 200 to 400 grams, and ten years to life with fines up to $100,000 above 400 grams (Texas Health and Safety Code § 481.1151, 2023). Texas House Bill 6 (2023) further enabled murder charges where a fentanyl-related sale results in death (Law Office of David D. White, 2024).

These penalties create implementation barriers throughout the county. Individuals with prior possession convictions face legal exposure from calling emergency services, possessing

naloxone in the context of drug use, or engaging with outreach workers. Felony drug records additionally affect employment, housing eligibility, and program enrollment in ways that compound the social determinants of substance use disorder.

### 4.7 Overdose Mapping and Data Sharing

Texas Senate Bill 1319, effective September 1, 2023, established mandatory overdose mapping and data sharing, requiring emergency medical services personnel, local health authorities, and law enforcement to report overdose incidents to a governmental entity maintaining a digital mapping system (Texas Senate, 2023). Dallas subsequently opted into the national Overdose Detection Mapping Application Program (Dallas Express, 2023). Two consequences follow. First, integration with law enforcement data systems raises a legitimate concern that people who use drugs may perceive the system as surveillance rather than public health infrastructure, potentially further deterring emergency calls. Second, access to the resulting incident data is restricted to credentialed public safety and public health agencies, placing the most current non-fatal overdose signal outside the reach of independent analysis, as documented in Section 7.5.

### 4.8 Insurance and Medicaid Policy

Texas is one of ten states that has not expanded Medicaid under the Affordable Care Act. The state uninsured rate exceeded 16% in 2023, the highest in the nation, and approximately 21% of Dallas County residents lack coverage (KFF, 2025; U.S. Census Bureau, 2026). Medicaid covers 47% of nonelderly adults with opioid use disorder nationally and 56% of those receiving medication for the condition (KFF, 2025). Expansion is associated with a 30% increase in the likelihood that low-income adults with substance use disorders receive treatment and a 6% lower opioid death rate in expansion states (Episcopal Health Foundation, n.d.; Texas Monthly,

2024). Non-expansion therefore functions as the single largest structural barrier to treatment access in the county.

### 4.9 Opioid Settlement Fund Distribution

Texas is projected to receive approximately $3.3 billion in opioid settlement funds over 18 years (Texas Comptroller of Public Accounts, n.d.), allocated 70% to the Texas Opioid Abatement Fund, 15% to the state, and 15% directly to counties and municipalities (National Academy for State Health Policy, 2025). Among 21 Texas localities analyzed by Rice University's Baker Institute, combined receipts totaled $60.8 million in fiscal years 2023 and 2024, with treatment programs the largest spending category at 27.6%, harm reduction accounting for 7.4% of committed funds, and 30.2% uncommitted (Baker Institute Drug Policy Program, 2025). Dallas County committed nearly $7 million to forensic toxicology upgrades from its allocation (Baker Institute Drug Policy Program, 2025).

**Table 2**

*Policy Summary Matrix: Dallas County Harm Reduction Landscape*

| Policy | Status (as of July 2026) | Analytical assessment |
|---|---|---|
| Naloxone standing order | Active (state) | Enabling. Authorizes broad access through a physician-issued framework; reaches 83.7% of audited pharmacies. Provides the legal basis for third-party storefront distribution. |
| Good Samaritan law | Severely limited (state) | Restrictive. The Jessica Sosa Act provides a narrow affirmative defense only, with an 18-month bar and single-use limitation; associated with deterred emergency calls. |

| Policy | Status (as of July 2026) | Analytical assessment |
|---|---|---|
| Fentanyl test strips | Died in committee | Restrictive. Passed the House 145–0 in 2025 but expired without a Senate committee hearing; remains classified as paraphernalia. |
| Syringe services programs | Unauthorized (state) | Restrictive. Only Bexar County is authorized; pending expansion legislation excludes Dallas County. |
| MAT environment | Partially accessible (federal/state) | Mixed. Prescriber pool expanded by DATA 2000 waiver elimination, but methadone still requires daily in-person attendance at certified programs. |
| Drug possession penalties | Highly punitive (state) | Restrictive. Fentanyl isolated under Penalty Group 1-B; murder charges available for fatal distribution. |
| Overdose mapping (SB 1319) | Active (state) | Mixed. Enables real-time incident mapping, but law enforcement integration raises community trust concerns and access is agency-restricted. |
| Medicaid non-expansion | Not expanded (state) | Restrictive. Largest structural barrier to treatment access; leaves approximately 21% of county residents uninsured. |
| Opioid settlement funds | Active allocation (local/state) | Underutilized. County committed approximately $7 million to forensic toxicology; statewide, harm reduction receives 7.4% of committed allocations. |

*Note.* Status and assessments reflect conditions as of July 2026. Sources: Baker Institute Drug Policy Program (2025); Dallas County Health and Human Services (2024); Texas Comptroller of Public Accounts (n.d.); Texas Legislature (2025); U.S. Government Accountability Office (2021).

## Section 5: Implementation Analysis

### 5.1 Naloxone Distribution: Funding, Delivery, and Measured Reach

Naloxone distribution in Dallas County flows through a fragmented multi-actor system rather than centralized infrastructure. At county level, DCHHS is the primary public distributor, providing free naloxone to organizations, shelters, libraries, restaurants, and bars upon training

completion (DCHHS, n.d.-a). At state level, Naloxone Texas serves as the primary supply backbone, distributing free intranasal naloxone to individuals and organizations (TXOTI, 2025).

The system has demonstrated significant fragility. The primary state distribution mechanism exhausted funding in 2022 amid a shortage of the lower-cost injectable formulation, forcing organizations to limit distribution to 48 doses per request (Be Well Texas, 2024). Delivery has been chronically unpredictable, with Dallas-Fort Worth organizations reporting sudden large deliveries following extended periods of small, delayed shipments (Be Well Texas, 2024). The federal funding dependency described in Section 4.1 compounds this fragility.

The DCHHS training-based model creates a structurally passive distribution framework that primarily reaches institutional partners that have proactively sought supply, rather than individuals at highest risk. Research on community distribution consistently finds that models requiring individuals to seek out services reach a fraction of the target population relative to embedded, proactive models (Fischer et al., 2025). Dallas Harm Reduction Aid, a community-based organization operating independently of DCHHS, distributes naloxone directly to people who use drugs through mobile delivery and peer outreach; as one of few entities in the county designed to engage this population, its capacity is heavily constrained relative to countywide need (NASEN Directory, 2024).

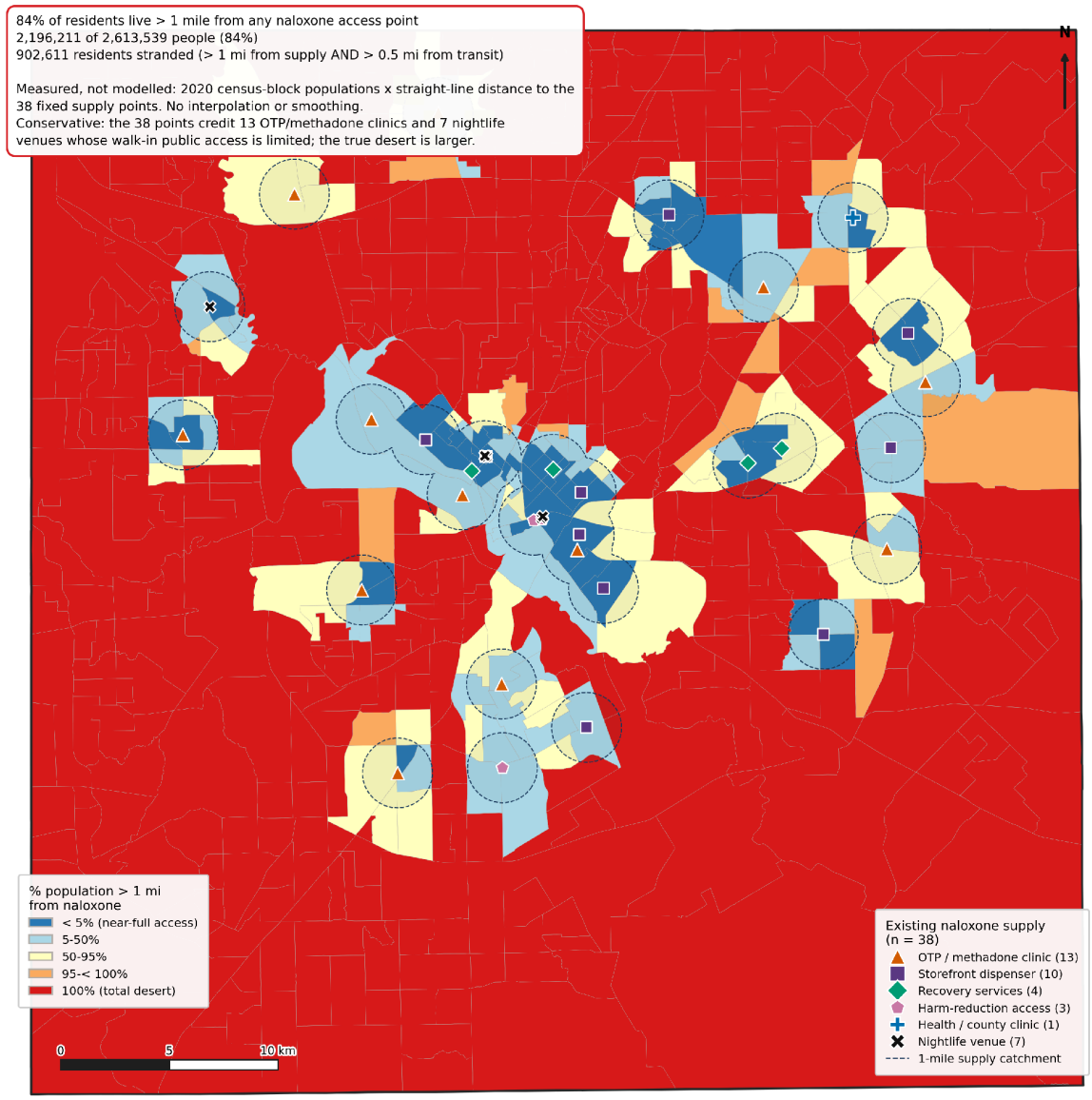
The geographic consequence of this configuration is measurable rather than inferred. Figure 1 presents the share of each tract's 2020 block population living more than one straight-line mile from any of the 38 fixed naloxone access points.

**Figure 1**

*The Measured Naloxone Access Desert in Dallas County*

**Figure 1. The measured naloxone access desert in Dallas County**

Share of each tract's 2020 population living more than 1 mile from any naloxone access point (straight-line, measured)




*Note.* Each of the 645 census tracts is shaded by the share of its 2020 census-block population living more than one mile (straight-line) from any of the 38 fixed naloxone access points, in five fixed classes (< 5%, 5–50%, 50–95%, 95–< 100%, and 100%, a total desert). Values are measured, not modeled: every block's population (2020 P.L. 94-171 count) is assigned to its nearest supply point in EPSG:32138, with no interpolation or smoothing. Point symbols mark the 38 fixed supply sites by category; one mobile harm reduction unit without fixed coordinates is omitted. The dashed outline is the union of 1-mile catchments around those sites, clipped to the county. Countywide, 2,196,211 residents (84.0%) live beyond one mile of supply and 902,611 are stranded (beyond one mile of supply and beyond one-half mile of transit). The measure is conservative: the 38 points credit 13 opioid treatment programs and 7 nightlife venues whose walk-in public access is limited, so the functional desert is larger. Sources: U.S. Census Bureau

2020 P.L. 94-171 block populations (P1_001N) and TIGER/Line 2024 tabulation blocks; Texas DSHS NarcanSites (April 2026); SAMHSA findtreatment.gov and verified naloxone inventory (2024–2026); county and tract boundaries from U.S. Census TIGER/Line 2024. Projection: NAD83 / Texas North Central (EPSG:32138). All layers are aggregated population counts or facility point locations; no individual-level data are used.

Against the exact 2020 decennial population of 2,613,539, the measured results are as follows. Beyond one mile of any fixed supply point: 2,196,211 residents (84.03%). Beyond two miles: 1,506,855 (57.66%). Beyond five miles: 394,283 (15.09%). Across 9,779 populated blocks, 902,611 residents meet all three stranded conditions, unable either to walk to naloxone or to readily reach it by transit. Tract-centroid distance to the nearest supply point averages 2.75 miles (median 2.34, maximum 10.2), with 16.6% of tracts within one mile and 43.4% within two miles.

Three tracts illustrate the extreme of the distribution, each entirely beyond one mile from supply and almost entirely stranded: tract 48113007601 (1,894 residents, 1,894 stranded), tract 48113007102 (5,441 residents, 4,885 stranded), and tract 48113007812 (3,212 residents, 3,072 stranded).

A note on comparability is warranted. Analyses in other jurisdictions have reported the proportion of overdose death locations falling within a short walk of supply. That metric is not computable for Dallas County, because no address-level or point-level mortality data exists publicly (Section 2.3). Three tract-centroid proxies were computed against a 500 m threshold: 1.71% of tracts (11 of 645) have a centroid within 500 m of supply; 2.43% when weighted by modeled overdose rate; and 1.42% of county population (37,003 residents) fall within 500 m. None of these is equivalent to a death-weighted figure. The defensible statement is that between roughly 1% and 2.5% of Dallas County falls within a 500 m walk of naloxone under any of these framings.

### 5.2 The Countywide 24-Hour Gap

Exactly one fixed naloxone access point in Dallas County operates on a 24-hour basis with usable coordinates: a fuel station in ZIP code 75210. Measured against that single point, 2,254,279 residents (86.25%) live more than five miles away. Tract-centroid distance to it averages 9.85 miles (median 10.07, maximum 20.7).

This finding carries an explicit caveat. One additional 24-hour harm reduction provider operates as a mobile unit without fixed coordinates and is therefore excluded from the distance computation. Were its routes geocoded, the measured 24-hour desert would shrink by an amount that cannot be estimated from available data. The finding is reported as a measured property of fixed infrastructure, not as a claim that no other after-hours access exists.

The temporal dimension compounds the geographic one. The DCHHS Overdose Prevention Hotline operates from 7 a.m. to 11 p.m. (CDC, 2025), leaving overnight hours, when overdose risk in nightlife and residential contexts is elevated, without either hotline support or, for the overwhelming majority of the county, proximate fixed supply.

### 5.3 Nightlife Corridor Supply

Existing venue-based dispensers cluster in two entertainment districts. In Deep Ellum (ZIP 75226), inventory includes two nightlife venues, a community center, and a storefront dispenser. In the Cedar Springs and Oak Lawn corridor (ZIP 75219), inventory includes four nightlife venues and a café.

Every nightlife-sited dispenser in the county inventory operates on non-24-hour access, and the Deep Ellum community center closes at 7 p.m., before the period of peak nightlife activity. Coverage across the county's remaining entertainment districts is thinner still: Lower

Greenville (ZIP 75206) has no inventory site, Uptown (ZIP 75204) has one, a recovery facility rather than a nightlife dispenser, and the Design District (ZIP 75207) has one.

These corridors constitute a category of risk that mortality-based prioritization structurally cannot detect, because the exposed population consists substantially of transient non-residents rather than residents whose deaths would be recorded to those geographies. Section 8.5 develops this point.

### 5.4 Language Access as a Measurable Absence

The demand-side case for Spanish-language service provision is well established. Dallas County is 41.3% Hispanic or Latino, and DCHHS reports the region's highest proportion of non-English-speaking households with limited English proficiency (DCHHS, 2024, p. 3). Seven of the ten highest-vulnerability ZIP codes identified in Section 8.3 are majority-Hispanic, with ZIP 75211 at 78.7%, 75217 at 71.5%, and 75212 at 67.9%. The 2024 demographic shift documented in Section 3.2, in which Hispanic males recorded the highest number of fentanyl-related deaths of any category, lands precisely in these geographies. Research on harm reduction utilization among racial and ethnic minority populations identifies language barriers among the most significant determinants of non-utilization (Ezell et al., 2024).

The supply-side finding is a documented absence. The assembled naloxone inventory carries no language attribute of any kind: available fields comprise name, address, city, ZIP, type, category, 24-hour status, hours, source, verification date, geocoding method, and mobile status. Neither the upstream state ArcGIS feed nor the federal treatment locator exposes a Spanish-service or bilingual-capacity flag. No Spanish-language service capacity is discoverable at any access point in the county, not because the analysis omitted the field, but because the field does not exist upstream. Language capacity is therefore unmeasurable from public data, which is

itself a finding about the state of harm reduction data infrastructure in a county where two in five residents are Hispanic or Latino.

### 5.5 Cultural Context of Drug Use

Effective implementation requires understanding how drug use is culturally situated across the county's communities, since these patterns determine which populations remain outside the reach of conventional outreach.

Dallas has a documented pattern of prescription opioid misuse predating the fentanyl era, which created the demand infrastructure into which fentanyl subsequently entered (D Magazine, 2021). Unlike heroin use, prescription pill misuse has been comparatively normalized across socioeconomic strata, particularly in suburban and middle-class communities, where it is often not perceived as drug use in the conventional sense. That normalization produced a substantial population of opioid-dependent individuals who do not self-identify as drug users and are systematically missed by outreach models targeting visibly marginalized populations.

Codeine-promethazine preparations consumed recreationally, known regionally as lean, are embedded in Southern hip hop culture and diffused across North Texas through association with music, youth identity, and social use rather than addiction (Ware et al., 2024). The pattern represents a specific blind spot: users frequently do not recognize codeine as an opioid, do not connect its use to overdose risk, and fall outside outreach designed around recognizable drug use patterns. The risk is compounded where the illicit supply substitutes fentanyl for codeine in counterfeit preparations.

The county's nightlife economy creates a polydrug context in which cocaine, MDMA, and counterfeit pills are consumed socially by individuals who do not identify as people who use drugs in any clinical sense. The 260% increase in opioid-stimulant combination deaths between

2016 and 2024 (DCHHS, 2024) reflects a supply in which fentanyl now crosses substance categories. The individuals at risk in these contexts are largely young, employed, and socially integrated. They do not present at harm reduction services and are not reached by outreach targeting encampments or recovery communities.

Drug use patterns are also linked to socioeconomic conditions. Poverty, unemployment, housing instability, trauma exposure, and limited economic opportunity are consistently identified as drivers of substance use disorder. A qualitative study of harm reduction practice across four Texas counties found implementation particularly difficult in communities where institutional distrust, rooted in experience with law enforcement, the criminal justice system, and discriminatory healthcare, creates barriers to engagement even where services are technically available (Claborn et al., 2023). The 2024 county needs assessment corroborated these dynamics locally, with focus groups and lived-experience interviews documenting stigmatizing policies, cross-sector collaboration deficits, and institutional stigma as barriers operating across the county's harm reduction landscape (Baumgartner & Ledat, 2024).

## Section 6: Comparative Case Study

Two cities serve as structured comparison cases. Houston functions as a structurally analogous case, demographically and legally comparable within the same restrictive state framework, illustrating the trajectory associated with maintaining the current approach. Philadelphia functions as a best-practice case, facing a more severe overdose crisis under conditions of comparable urban poverty and demographic complexity, and having implemented evidence-based harm reduction at scale with documented mortality decline. As noted in Section 2.7, these cases are not controlled conditions and the comparison is offered as structured qualitative evidence rather than causal inference.

### 6.1 Houston, Texas: Structurally Analogous Case

Harris County holds approximately 4.8 million residents. Like Dallas, Houston is a majority-minority city with a large Hispanic population, a significant Black community, concentrated urban poverty, and a sprawling geographic footprint that complicates centralized service delivery. Harris County recorded 307 opioid overdose deaths in 2019, compared with Dallas County's 163 that year, a difference broadly proportional to the two counties' populations (Claborn et al., 2023). By 2022, substance-involved deaths in Harris County had risen to 1,177 from 673 in 2018, a 75% increase across four years (Houston Public Media, 2024). In 2023, more than 600 Harris County residents died of opioid overdose, an average of 11 per week (Houston Landing, 2025).

Houston operates under identical state constraints: no comprehensive Good Samaritan law, no authorized syringe services program, criminalized fentanyl test strips, Medicaid non-expansion, and the same possession penalty structure. Houston is not a city that attempted a different approach and failed; it is a city constrained, as Dallas is, to operating within the same restrictive framework.

Implementation has additionally been characterized by limited transparency and diffuse targeting despite $4.4 million in settlement receipts. Baker Institute analysis found that 17% of Houston's 155 retail pharmacies did not carry naloxone as of 2023, that only 14 state-licensed opioid treatment programs in Harris County provided methadone, and that only 67% of surveyed Texas pharmacies stocked buprenorphine (Baker Institute Drug Policy Program, 2025; Houston Landing, 2025). Populations most affected mirror those in Dallas: Black and White non-Hispanic community members, people experiencing homelessness, and justice-involved individuals (Houston Public Media, 2024).

Houston demonstrates the outcome of insufficient tools operating within an insufficient policy framework. Naloxone distribution without syringe access, without functional Good Samaritan protection, without legal drug checking, and without treatment access parity produces incremental results at best.

### 6.2 Philadelphia, Pennsylvania: Best-Practice Case

Philadelphia holds approximately 1.6 million residents with a demographic composition of roughly 39% Black, 35% White, and 15% Hispanic, a majority-minority profile comparable to Dallas with similar entrenched economic inequality (World Population Review, 2026). Its poverty rate of approximately 20% to 22% exceeds Dallas County's, and the city has consistently ranked among the poorest large cities in the United States (Philadelphia Inquirer, 2024). Philadelphia recorded 1,413 overdose fatalities in 2022, its highest on record.

Pennsylvania's framework contrasts sharply with Texas's. Act 139 (2014) provides comprehensive Good Samaritan immunity from prosecution for simple possession for individuals who call emergency services during an overdose (NEXT Distro, n.d.). Fentanyl test strips are legal and actively distributed, with the city health department distributing more than 180,000 fentanyl test strips and nearly 50,000 xylazine test strips in 2023 alone (Philadelphia Department of Public Health, 2025). Syringe services are authorized and operational, with Prevention Point Philadelphia providing syringe access, naloxone, wound care, testing, overdose prevention training, and treatment linkage for over 30 years (NEXT Distro, n.d.).

Philadelphia's distribution infrastructure is notably not limited to institutional access points. Naloxone is available over the counter, by mail, at 61 fire stations through a targeted initiative designed to reach Black Philadelphians, and through 24-hour vending machines in high-need neighborhoods (Substance Use Philly, n.d.). The city conducted door-to-door outreach

across 100,000 doors in high-overdose neighborhoods, with up to 88% of residents who answered accepting some form of resource (Philadelphia Inquirer, 2025). In 2023 the city conducted more than 230 harm reduction trainings and distributed over 100,000 naloxone doses, nearly double its 2022 volume (Philadelphia Department of Public Health, 2025).

Outcomes followed. Philadelphia recorded its first year-over-year decline since 2018 in 2023, falling approximately 7% from the 2022 peak of 1,413 to 1,315 deaths (Philadelphia Department of Public Health, 2025; WHYY, 2024). By 2024, deaths had fallen to 1,069, a 24% decline from the 2022 peak and an 18% decrease from 2023 (Philadelphia Department of Public Health, 2026). Deaths among Black Philadelphians decreased 27% between 2023 and 2024, reversing a multi-year trend of disproportionate increase (Philadelphia Department of Public Health, 2026). Pennsylvania statewide recorded a 29% decline between 2023 and 2024 (Philadelphia Inquirer, 2025). Fentanyl remained detected in 77% of 2024 deaths (Philadelphia Department of Public Health, 2026), indicating that the decline did not result from a change in supply composition.

### 6.3 Transferable Lessons

The comparison supports a specific conclusion: the variables most strongly associated with divergent outcomes across these three jurisdictions are policy permissiveness and implementation scale rather than demographic complexity, poverty concentration, or supply severity.

Three elements of the Philadelphia approach are transferable within the current Texas legal environment. First, proactive neighborhood-level outreach requires no policy change and produced acceptance rates as high as 88% in Philadelphia. Second, 24-hour access points embedded in existing neighborhood infrastructure reached populations that institutional

distribution models consistently miss, a finding of direct relevance given that Dallas County currently maintains one fixed 24-hour access point (Section 5.2). Third, directing resources to trusted community messengers rather than exclusively to government institutions expanded reach among populations with justified institutional distrust.

The underlying mechanism is consistent across the Philadelphia evidence: placing naloxone in non-clinical, everyday locations reaches recreational users, individuals who do not self-identify as having a substance use disorder, bystanders in high-risk social environments, and residents of high-need neighborhoods lacking transportation or service awareness. Dallas County's current distribution infrastructure has not meaningfully adopted this model, and Section 5.1 quantifies the consequence.

## Section 7: Gap Identification

The gaps documented here form a system of compounding barriers in which each unaddressed gap amplifies the effect of others.

### 7.1 Policy Gaps

Texas remains one of three states without comprehensive Good Samaritan protection, and the Jessica Sosa Act's narrow affirmative defense provides no meaningful protection for the populations most likely to witness an overdose (Claborn et al., 2023; Section 4.2). Fentanyl test strips remain criminalized despite unanimous House support for legalization (Section 4.3), a barrier of particular consequence given the substance profile in Section 3.2. Dallas County has no authorized syringe services program and is excluded from proposed expansion (Filter Magazine, 2025). Medicaid non-expansion leaves approximately 21% of residents without the coverage that nationally funds 47% of treatment for opioid use disorder (KFF, 2025).

### 7.2 Implementation Failures

Distribution remains structurally passive, reaching institutional partners that seek supply rather than individuals at highest risk (Section 5.1). The 75% increase in hotline call volume between November 2024 and March 2025 indicates demand outpacing capacity (CDC, 2025). The county operates no peer navigator program embedded in emergency departments, despite the post-overdose encounter representing a high-leverage window for treatment linkage, a gap confirmed by the 2024 needs assessment (Baumgartner & Ledat, 2024). Treatment capacity for individuals under 18 is absent within the county, and rehabilitation capacity generally lags demand (Dallas Express, 2024).

### 7.3 Geographic Disparities

Section 5.1 establishes the geographic gap as measured rather than inferred: 84.03% of residents live beyond one mile of supply, and 902,611 are stranded from both supply and transit. Existing infrastructure is concentrated in institutional settings distributed inequitably relative to need, and pharmacies in higher-poverty neighborhoods are consistently less likely to stock naloxone, producing an inverse relationship between need and access that mirrors national patterns (Fischer et al., 2025). Section 8 identifies the specific geographies where this convergence is most acute.

### 7.4 Demographic Disparities

Burden is not evenly distributed. Overall mortality is highest among males, White individuals, and adults aged 35 to 64, while cocaine-related deaths concentrate among Black males aged 55 to 64 and the 2024 fentanyl burden shifted to Hispanic males (DCHHS, 2024; Section 3.2). Research on harm reduction utilization among racial and ethnic minority populations identifies four consistent barriers: limited awareness of resources, distrust of their quality, lower resource concentration in minority communities, and internalized stigma (Ezell et

al., 2024). Section 5.4 documents the language access dimension as an unmeasurable absence in the supply data itself.

Youth represent an underserved population, with counterfeit prescription pills distributed through social media platforms creating a risk pathway outside programs designed around adult use patterns. Justice-involved individuals represent a second underserved group, with the post-release period among the highest-risk windows for fatal overdose as tolerance declines during incarceration while access resumes upon release (Al-Awadhi & Vegiraju, 2026).

### 7.5 Data Gaps

The suppression regime documented in Section 2.3 is itself a structural gap. Death counts are suppressed in 592 of 645 tracts and 138 of 150 native reporting units, and no public ZIP-level mortality product exists. Nationally, overdose deaths are acknowledged to be undercounted relative to actual incidence, as deaths occurring outside medical settings or misclassified on death certificates escape mortality surveillance (Ruhm, 2018). Non-fatal events are undercaptured at a still greater rate (Section 3.3).

Three specific datasets would materially improve prioritization and are unavailable. Real-time suspected overdose incident data is restricted to credentialed agencies (Section 4.7). Case-level medical examiner records, which would enable death-point access analysis and residence-versus-incident concordance, are available only by request. Emergency medical services naloxone administration data is not published, removing a demand proxy independent of mortality. Treatment enrollment and retention data are not published at county level, preventing assessment of treatment system performance.

### 7.6 Access Barriers

Cost remains a barrier for the uninsured 21% of residents, and while naloxone is available at no cost through public programs, medication treatment requires ongoing clinical interaction carrying cost implications. Transportation is a substantial barrier in a county of this geographic extent, and Section 5.1 quantifies it: 902,611 residents lack both proximate supply and proximate transit. Hours of operation constitute a third barrier, with one fixed 24-hour access point countywide (Section 5.2) and hotline support ending at 11 p.m. Language access is documented in Section 5.4.

### 7.7 Cultural and Community Barriers

Stigma operates at public, institutional, and internalized levels and functions as a barrier at every point in the care continuum (McGinty & Barry, 2020). Individuals who use drugs are routinely labeled in ways that rationalize institutional mistreatment, and anticipated stigmatization leads many to conceal use, use alone, and avoid emergency calls (Febres-Cordero et al., 2023). The dynamic is acute in communities with documented experience of discriminatory enforcement, where criminalization has historically meant that calling for help carries arrest risk. The county needs assessment identified stigma as a consistent barrier operating across both service providers and people who use drugs, and recommended internal staff education on stigma and trauma-informed practice within organizations serving people with lived experience (Baumgartner & Ledat, 2024).

The cultural patterns documented in Section 5.5 compound this: normalized pill use across socioeconomic strata, codeine preparations consumed as social rather than addictive practice, and polydrug nightlife consumption together produce substantial populations at fentanyl risk who neither identify with harm reduction services nor recognize their own exposure.

### 7.8 Structural Limitations

Several gaps are not amenable to local solution absent state action. The absence of syringe services authorization is a state prohibition no local entity can circumvent. The weakness of Good Samaritan protection is a legislative product that programmatic innovation cannot address. Medicaid non-expansion removes the largest funding mechanism for treatment from local availability. The federal funding dependency underlying more than 90% of the county harm reduction budget is a fragility that no local reallocation fully resolves. These limitations argue for a multi-level strategy pursuing what is achievable within the current environment while building the evidence base necessary to change what is not.

## Section 8: Geographic Prioritization

Resources deployed in the wrong location produce marginal outcomes regardless of program quality. This section identifies where the convergence of measured access deficit, socioeconomic vulnerability, and modeled mortality most strongly supports prioritized intervention, and closes by presenting mortality at the finest resolution the suppression regime permits.

### 8.1 Composite Vulnerability Tiers

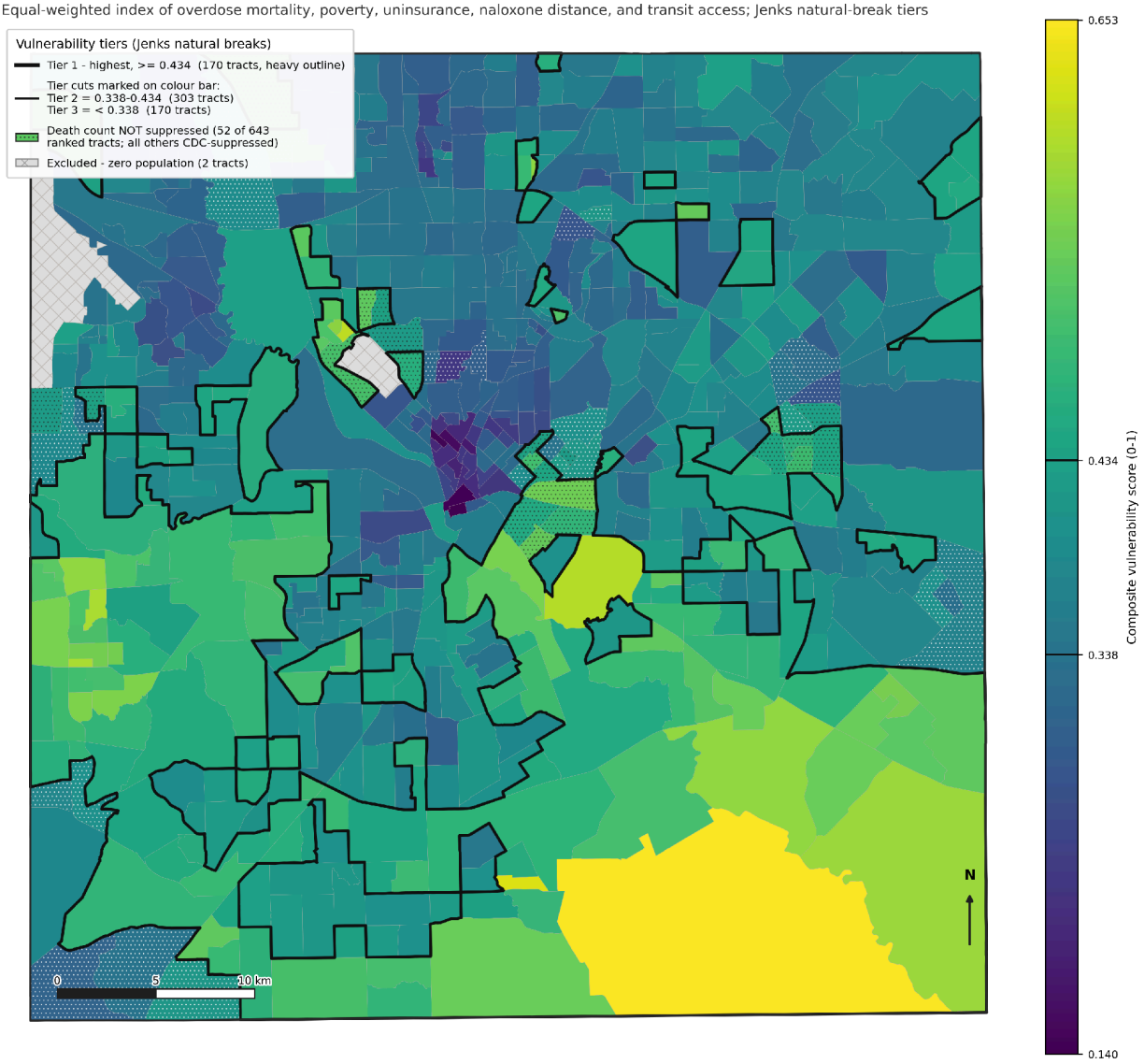
Figure 2 presents the composite index described in Section 2.5.

**Figure 2**

*Composite Overdose Vulnerability Index for Dallas County Census Tracts*

**Figure 2. Composite overdose-vulnerability index, Dallas County**

Equal-weighted index of overdose mortality, poverty, uninsurance, naloxone distance, and transit access; Jenks natural-break tiers




*Note.* Fill color represents a composite vulnerability score, the equal-weighted mean (0.20 each) of five min-max-normalized layers: model-based overdose mortality rate, poverty rate (ACS 2020–2024), uninsured rate (ACS 2020–2024), measured distance to the nearest naloxone access point, and inverted transit connectivity (DART stops within 0.5 mi). Of 645 tracts, 643 are ranked. Tracts fall into three Jenks natural-break tiers (Tier 3 to Tier 2 cut = 0.338; Tier 2 to Tier 1 cut = 0.434); Tier 1 (highest vulnerability, 170 tracts) is outlined in heavy black. Fine dotted marking identifies the 52 of 643 ranked tracts whose underlying death counts are not CDC-suppressed; every other ranked tract carries a suppressed count (592 of 645 tracts countywide, 591 of them among the ranked set). Two zero-population tracts are cross-hatched and excluded. Sensitivity analysis re-running the full pipeline with re-weighted layers shows the tiering is robust to mortality weighting but sensitive to distance weighting: doubling the mortality weight leaves 86.9% of tracts in the same tier, while doubling the distance weight relocates 21.5%. All relocations occurred between adjacent tiers, with no Tier 1 to Tier 3

movement under either variant. Sources: CDC NCHS model-based drug overdose death rates (4day-mt2f), 12-month period January–December 2025, data as of July 22, 2026; ACS 2020–2024 5-year (S1701 poverty; S2701 uninsured); CDC/ATSDR SVI 2022; Texas DSHS NarcanSites (April 2026) and verified naloxone inventory; DART GTFS (July 2026); boundaries from U.S. Census TIGER/Line 2024. Projection: EPSG:32138. All layers are tract aggregates; no individual-level data are used.

Composite scores range from 0.140 to 0.653. Tier assignment yields 170 tracts in Tier 1, 303 in Tier 2, and 170 in Tier 3. Under the distance-doubled sensitivity variant, tier counts shift to 130, 333, and 180; under the mortality-doubled variant, to 159, 281, and 203.

### 8.2 What Drives the Tier 1 Designation

Decomposing the Tier 1 aggregate across its 170 tracts reveals that the designation is driven first by transit inaccessibility, which contributes 38.2% of the mean composite, followed by uninsurance at 17.2%, modeled mortality at 15.2%, distance to supply at 15.2%, and poverty at 14.2%.

This decomposition requires an explicit interpretive caution. Tier 1 does not mean highest death rate. It means highest convergence of vulnerability across five dimensions, of which mortality is one and among the smaller contributors. Reinforcing that caution, 149 of the 170 Tier 1 tracts carry CDC-suppressed underlying counts, so their mortality inputs are model-based estimates rather than observed values. Readers and users of this analysis should not interpret Tier 1 membership as an epidemiological claim about where the most deaths occur. It is a claim about where structural barriers to accessing overdose prevention resources are most densely stacked.

### 8.3 Relationship to Prior ZIP-Level Framing

Because prior analyses of the county described priority geography in ZIP terms, the tract-level tiers were crosswalked to ZIP Code Tabulation Areas using the Census 2020 ZCTA-to-tract relationship file with land-area apportionment, with the modal tier defined as the tier holding the largest population share (Table 3).

**Table 3**

*Composite Vulnerability Score by ZIP Code Tabulation Area*

| ZIP | Modal tier | Mean composite | Tracts |
|---|---|---|---|
| 75217 | 1 | 0.4713 | 22 |
| 75210 | 1 | 0.4674 | 4 |
| 75215 | 2 | 0.4628 | 6 |
| 75203 | 2 | 0.4360 | 10 |
| 75227 | 2 | 0.4100 | 13 |
| 75224 | 2 | 0.4078 | 11 |
| 75211 | 2 | 0.4045 | 19 |
| 75212 | 2 | 0.3916 | 7 |
| 75226 | 2 | 0.3532 | 5 |
| 75204 | 3 | 0.2573 | 15 |
| 75201 | 3 | 0.2325 | 9 |

The core high-vulnerability geography identified in prior work holds: all six South and West Dallas ZIP codes previously designated highest priority remain within the Tier 1 to Tier 2 band, and none falls to Tier 3. Exact labels shift because this composite incorporates measured naloxone distance and transit connectivity, layers absent from deprivation-only analyses. ZIP 75217 is promoted from a prior secondary designation to the single highest-scoring ZIP in the county, and 75227 rises correspondingly, while the affluent central ZIP codes 75201 and 75204 fall to Tier 3. An independent check supports the promotion: DCHHS identifies 75217, 75216, and 75215 as the ZIP codes with the highest resident overdose death counts across 2016 to 2024 (DCHHS, 2024, Fig. 9); of these, 75217 and 75215 rank first and third in the present ordering,

while 75216 does not place among the highest-scoring ZIP codes under this composite, consistent with the index measuring converged structural barriers rather than mortality alone (Section 8.2).

The crosswalk also resolves an ambiguity present in prior tiering, in which ZIP 75212 was assigned to two tiers simultaneously. Under modal population-share mapping, each unit receives exactly one tier.

### 8.4 Priority Geographies

**South Dallas (75215, 75210).** These ZIP codes rank among the county's highest across poverty, housing instability, and public health vulnerability. The CDC/ATSDR index ranks Dallas County as more vulnerable than approximately 80% of Texas counties, with vulnerability concentrated in the indicators most pronounced in South Dallas's predominantly Black residential population (ATSDR, 2022; Texas Community Health News, 2020). ZIP 75210 additionally hosts the county's only fixed 24-hour access point, which paradoxically makes it both a priority geography and the reference point against which the countywide 24-hour desert in Section 5.2 is measured.

**Oak Cliff and West Dallas (75211, 75212, 75203, 75224).** A contiguous high-need zone west and southwest of the central business district. Oak Cliff is the county's most populous Hispanic-majority area, with concentrated poverty, elevated uninsurance, and the language access gap documented in Section 5.4. West Dallas, in the Trinity River floodplain, carries a historical pattern of environmental and public health underinvestment. Dense commercial corridors along Jefferson Boulevard, Zang Boulevard, Davis Street, and Singleton Boulevard provide existing foot traffic infrastructure relevant to embedded distribution.

**Pleasant Grove and Southeast Dallas (75217, 75227).** ZIP 75217 records the highest composite score of any ZIP in the county. The area is separated from central service infrastructure by the Great Trinity Forest and has limited transit connectivity, so residents face both distance and transportation barriers. Childhood poverty reaches 38.9% in 75217 and 24.6% in adjacent 75227 (U.S. Census Bureau, 2026). Overdose events here are less likely to generate incident data, less likely to result in emergency calls absent Good Samaritan protection, and more likely to fall within suppressed counts.

**Northwest corridors (75220, 75231).** These geographies do not appear in prior deprivation-based framings but emerge prominently in the placement analysis in Section 8.6, because they combine high unserved population density with elevated vulnerability. Their emergence is a direct consequence of incorporating measured access into the prioritization.

### 8.5 Zones of Structurally Undetectable Risk

Not all high-risk areas are visible in mortality data, and two categories warrant separate treatment.

The nightlife and entertainment corridors documented in Section 5.3, including Deep Ellum, Uptown, Lower Greenville, and the Design District, generate risk through polydrug consumption in social settings, with fentanyl-adulterated cocaine and counterfeit pills as primary vectors. Events in these contexts are more likely to be managed by bystanders without emergency contact, given social stigma and the legal exposure created by inadequate Good Samaritan protection. National literature documents that 42.6% of fatal overdoses between October 2020 and March 2024 occurred with a potential bystander present (Brookings Institution, 2025), a proportion plausibly higher in nightlife contexts. Critically, the exposed population is substantially non-resident, so deaths are attributed to residence geographies

elsewhere and the corridor's risk cannot register in residence-based mortality mapping regardless of suppression.

Subsidized housing concentrations across southern and western Dallas constitute a second category, where elevated risk is driven by poverty, social isolation, and co-occurring conditions, and where small-cell suppression is most likely to obscure the local signal.

### 8.6 Coverage-Maximizing Placement Analysis

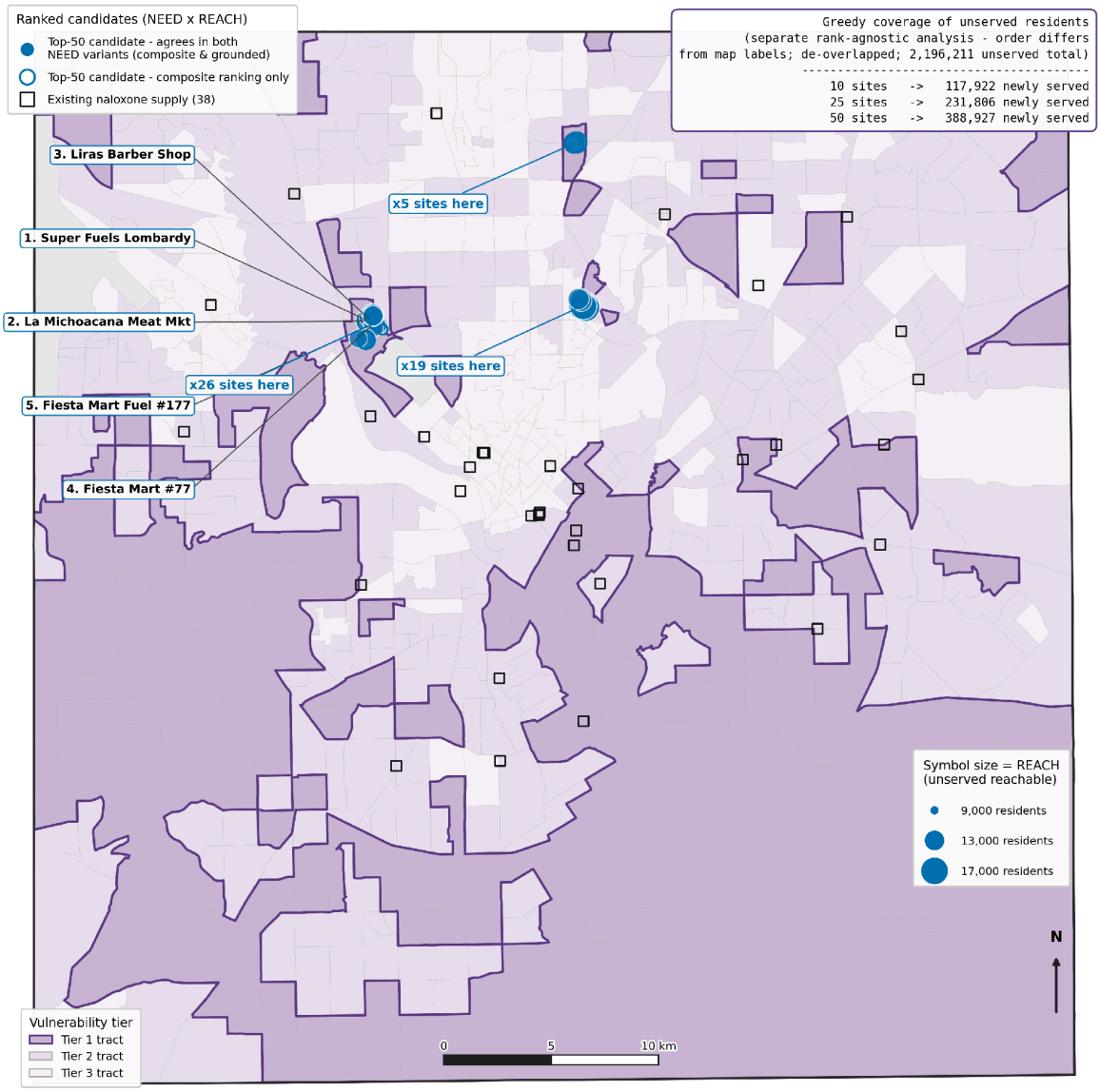
Because mortality is suppressed at fine geography while the population lacking access is exactly countable, coverage of currently unserved residents is the more defensible optimization target. Figure 3 presents the ranked candidate analysis.

**Figure 3**

*Ranked Naloxone Placement Strategy*

**Figure 3. Ranked naloxone placement strategy**

Top 50 of 5,428 candidate storefronts by NEED x REACH (composite rank), sized by unserved residents reachable; vulnerability tiers shaded




*Note.* Candidate storefronts are ranked by need multiplied by reach. Reach is the number of currently unserved residents (2020 census blocks beyond one mile of existing supply) within a half-mile walk of a candidate; need is tract vulnerability. Two need variants are computed, the five-layer composite index (Figure 2) and a mortality-free grounded index (poverty, uninsurance, Social Vulnerability Index, and measured naloxone distance), each multiplied by normalized reach. The two rankings agree closely (Spearman rho = 0.966; 45 of the top 50 shared). The map shows the 50 top-ranked candidates by composite score, sized by reach; solid symbols agree in both variants, hollow symbols rank in the composite variant only. Hollow squares mark the 38 existing supply sites; muted shading shows vulnerability tiers. The five highest-ranked candidates are labeled. The inset reports a separate rank-agnostic greedy coverage analysis whose ordering differs from the map labels: the best 10, 25, and 50 non-overlapping sites would newly reach 117,922, 231,806, and 388,927 of the 2,196,211 unserved residents. Candidate

categories derive from license class and point-of-interest tags rather than field verification. Sources: Dallas County business licenses, TABC off-premise permits, and OpenStreetMap points of interest (July 2026); U.S. Census 2020 P.L. 94-171 block populations and TIGER/Line 2024 tabulation blocks; CDC NCHS 4day-mt2f; ACS 2020–2024 5-year; CDC/ATSDR SVI 2022; Texas DSHS NarcanSites (April 2026) and verified naloxone inventory. Projection: EPSG:32138. All population and vulnerability inputs are aggregated; no individual-level data are used.

The greedy coverage analysis produces the study's central actionable finding. Ten optimally selected non-overlapping sites would newly place supply within a half-mile walk of 117,922 currently unserved residents (5.4% of the unserved population). Twenty-five sites would reach 231,806 residents (10.6%). Fifty sites would reach 388,927 (17.7%). The steepness of the initial gains reflects the concentration of unserved population density in a small number of corridors.

Two clusters dominate. The Lombardy Lane and Webb Chapel corridor in ZIP 75220 (tracts 48113007205, 48113007206, and 48113009804) holds composite ranks 1 through 5 and several category leaders. A second cluster in the Park Lane and Vickery Meadow area of ZIP 75231 (tract 48113007815) contains the single highest-reach candidate in the county, a grocery store adjacent to 17,437 unserved residents, along with a public library branch adjacent to 16,110. These two pockets combine high unserved population density with elevated vulnerability and represent the strongest coverage-maximizing targets identified.

Recommended host types are ordinary neighborhood retail: fuel stations, corner and convenience stores, off-premise beverage retailers, laundromats, salons and barbershops, groceries, and public library branches. The categorization caveat in Section 2.6 applies: license-derived categories require field verification before siting.

The close agreement between the two need variants (Spearman rho = 0.9659) merits emphasis as a methodological result. Because the grounded variant contains no mortality input whatsoever and produces nearly the same priority ordering as the composite variant, the

placement recommendation does not depend on the suppressed, model-based mortality layer. In a jurisdiction where mortality is the least reliable available signal, that independence is a substantive strength rather than a technical footnote.

### 8.7 Mortality Geography at Its Honest Resolution

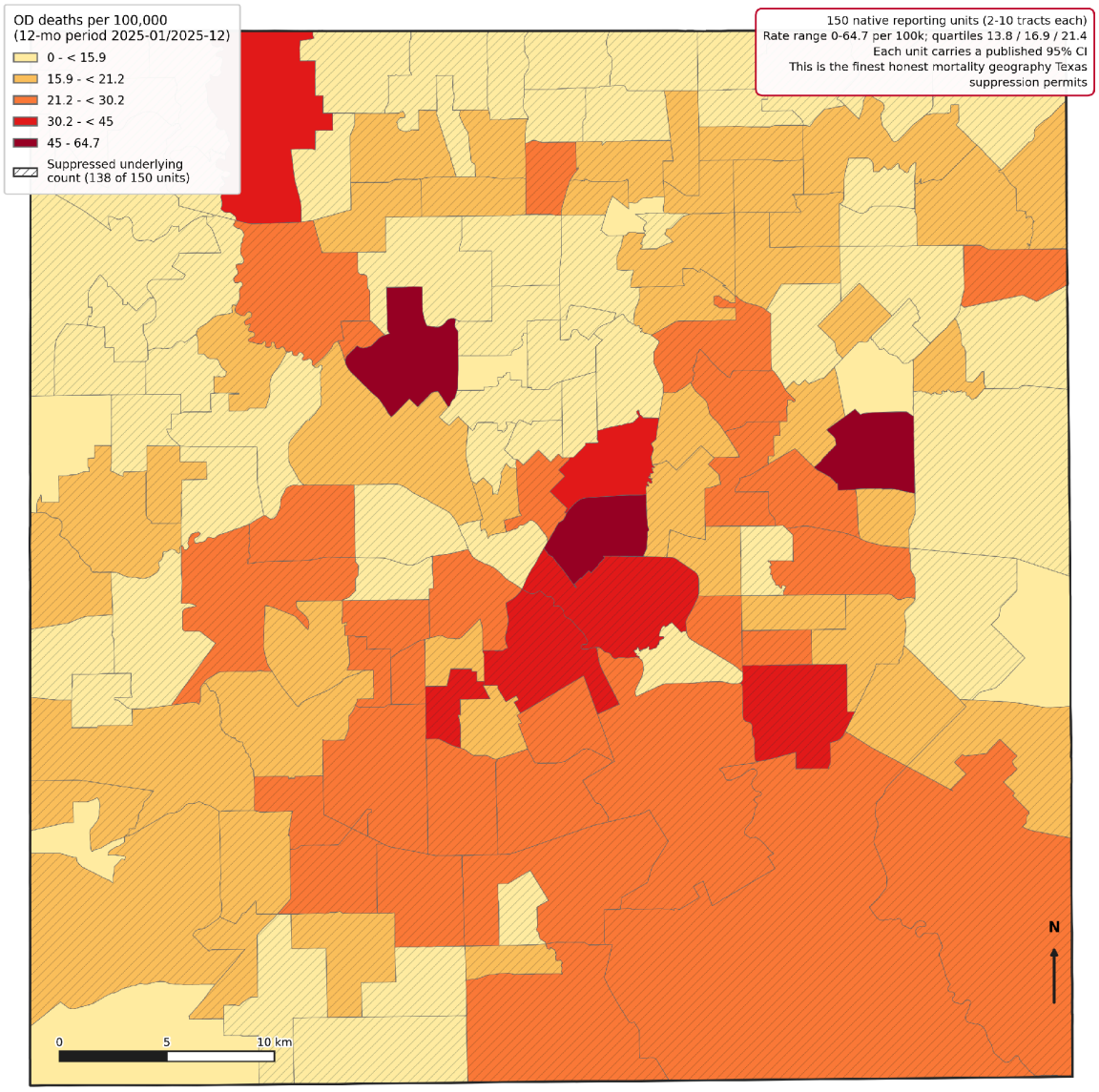
Figure 4 presents overdose mortality at the finest geography the suppression regime permits without fabricating counts, and serves as a check on the preceding analysis.

**Figure 4**

*Overdose Mortality at CDC Native Reporting Units, Dallas County*

**Figure 4. Overdose mortality at CDC native reporting units**

Model-based overdose death rate per 100,000 across the 150 CDC aggregation units covering Dallas County (Jenks natural breaks)




*Note.* Each of the 150 CDC aggregation units (unions of 2 to 10 census tracts) is shaded by its model-based overdose death rate (deaths per 100,000, 12-month period ending December 2025), classified into five Jenks natural-break ranges. Rates span 0 to 64.7 per 100,000, with quartiles at 13.8, 16.9, and 21.4. Diagonal hatching marks the 138 units (92%) whose underlying death counts are CDC-suppressed. Texas mortality suppression prevents publication of raw tract-level counts, so the tract product releases model-based rates aggregated to these native units; a 95% confidence interval is published for each unit. Sources: CDC NCHS model-based drug overdose death rates (4day-mt2f), 12-month period January–December 2025, data as of July 22, 2026; boundaries from U.S. Census TIGER/Line 2024. Projection: EPSG:32138. No individual-level death records exist in the source or in the analytic repository.

Across the 150 native units, rates distribute with a minimum of 0.0, first quartile of 13.75, median of 16.9, third quartile of 21.425, mean of 18.126, and maximum of 64.7 per 100,000. Seven units are published at an exact modeled rate of zero.

Two properties of this figure should govern its interpretation. First, 92% of units carry suppressed underlying counts, so the shading represents smoothed model output rather than observed death density, and every unit carries a published confidence interval that a single choropleth color cannot convey. Second, this is the finest mortality geography that exists publicly for Dallas County. Any map presenting overdose mortality at tract resolution for this county, including the mortality layer within Figure 2, has disaggregated these units and is displaying variance the source data does not resolve.

Figure 4 is therefore included not as a prioritization instrument but as a statement of the evidentiary limit. It is the boundary of what can be known about where people are dying in this county from public data, and the distance between that boundary and the block-level precision of Figure 1 is the central methodological argument of this paper.

## Section 9: Metrics and Evaluation

The measured framework produces its own evaluation instruments, which is a practical advantage over prioritization schemes resting on suppressed data. Four short-term metrics are directly recomputable from public data on the analytic pipeline's refresh cadence, and each connects to a gap identified in Section 7.

**Population beyond one mile from supply** measures the geographic access gap (Section 7.3) and is the headline indicator. Its baseline is 2,196,211 residents (84.03%). Any placement, closure, or inventory change recomputes this figure exactly.

**Stranded population** measures the compounded access and transportation barrier (Sections 7.3 and 7.6), with a baseline of 902,611 residents across 9,779 blocks. This metric is more sensitive than the first to placements in transit-poor geographies and therefore distinguishes between placements that are geographically equivalent but functionally different.

**Fixed 24-hour access points and population beyond five miles of them** measures the temporal access gap (Sections 5.2 and 7.6), with a baseline of one point and 2,254,279 residents (86.25%). Because the baseline is a single site, this metric is unusually responsive: each additional 24-hour location produces a large, measurable, and reportable change.

**Population within a 500 m walk of supply** measures the walkable access gap, with a baseline of 37,003 residents (1.42%). This is the strictest of the four and the most appropriate for evaluating dense-corridor placement specifically.

Long-term evaluation necessarily relies on mortality surveillance despite its limitations. The VSRR county series provides a 12-month-ending trend updated monthly, with the caveat that recent points revise upward. The DCHHS annual surveillance report provides finalized calendar-year counts with demographic and substance-specific breakdowns, permitting assessment of whether the disparities identified in Section 7.4, particularly the 2024 shift in fentanyl burden toward Hispanic males, narrow or widen.

Two evaluation limits should be stated plainly. Attribution of any mortality change to any specific distribution intervention is not supportable from these data, and no such attribution should be inferred. And the access metrics measure supply proximity rather than naloxone reaching individuals, which would require distribution volume and utilization data that no current public source provides.

## Section 10: Recommendations for Policy and Practice

Four recommendations follow from the preceding analysis. Each is tied to a specific gap finding and, where geographic, to a specific mapped location. All are addressed to public health authorities, community-based organizations, and legislative bodies generally rather than to any single organization.

**First, coverage-maximizing placement of naloxone in everyday commercial and public settings within the highest-yield corridors.** This addresses the measured access gap in Sections 5.1 and 7.3, and targets the Lombardy Lane and Webb Chapel corridor in ZIP 75220 and the Park Lane and Vickery Meadow area in ZIP 75231 identified in Section 8.6, alongside the Tier 1 geographies in Section 8.4. The evidence base for non-institutional placement is established: a naloxone vending machine study conducted from October 2023 to May 2024 found that 46.2% of recipients reported they would not otherwise have had access, directly demonstrating the population reach advantage of non-institutional siting (Dahlem et al., 2025). Removing registration requirements and product limits significantly improves utilization among minoritized and unstably housed populations, precisely the groups least served by the current institutional model (Vadiei et al., 2025). This intervention requires no policy change: it is permissible under the existing Texas standing order framework described in Section 4.1. Feasibility is constrained by the need for field verification of candidate sites (Section 2.6) and by host willingness, neither of which this analysis can assess.

**Second, expansion of fixed 24-hour access.** This addresses the temporal gap in Sections 5.2 and 7.6, where the baseline is one fixed 24-hour point serving a county in which 86.25% of residents live more than five miles away. Because that baseline is a single site, the marginal return on each additional 24-hour location is exceptionally high by the measured metrics in Section 9. Candidate hosts already operating continuously, including fuel stations and 24-hour

retail, are concentrated in the corridors identified in Section 8.6 and in the nightlife districts in Section 5.3 where every current dispenser closes before peak risk hours.

**Third, linguistic and cultural competency built into service design and into the data infrastructure itself.** This addresses the demographic and access gaps in Sections 5.4, 7.4, and 7.6. The demand-side case is established by a 41.3% Hispanic or Latino population, the region's highest concentration of limited-English-proficiency households, and the 2024 shift placing Hispanic males as the leading fentanyl-death group. The supply-side finding is that no Spanish-language service capacity is discoverable at any access point because no upstream dataset records the attribute. Two distinct actions follow. Programmatically, evidence indicates that interventions co-designed with community members produce more sustained engagement than programs developed externally and translated afterward (Claborn et al., 2023), and that language barriers rank among the strongest determinants of non-utilization (Ezell et al., 2024). Structurally, adding a language capacity field to public naloxone site inventories would make the gap measurable, which is a precondition for closing it. Outreach messaging should additionally account for the cultural patterns in Section 5.5, since framing built around self-identified drug use will not reach populations who do not so identify.

**Fourth, legislative action on the structural barriers that local intervention cannot address.** This addresses Section 7.8. The comparative evidence in Section 6 indicates that jurisdictions implementing the full evidence-based toolkit produce measurably better outcomes independent of demographic and socioeconomic conditions. Within the current legislative environment, fentanyl test strip decriminalization represents the most immediately achievable target, having passed the House 145 to 0 in 2025 with failure occurring through procedural non-action in Senate committee rather than substantive opposition (Section 4.3). Replacement of

the Jessica Sosa Act's restricted affirmative defense with genuine immunity, modeled on Pennsylvania's Act 139, follows as the next priority given the documented relationship between call deterrence and preventable mortality (Sections 4.2 and 6.2). Syringe services authorization inclusive of Dallas County and Medicaid expansion represent longer-term structural reforms requiring sustained coalition-building. Additionally, the settlement fund allocation documented in Section 4.9, in which harm reduction receives 7.4% of committed funds against 27.6% for treatment and 30.2% uncommitted, represents the most significant near-term funding mechanism available for the distribution infrastructure whose absence this analysis measures.

## Section 11: Conclusion

The central finding of this analysis is that Dallas County's naloxone access deficit is not an inference. Against an exact decennial count of 2,613,539 residents, 2,196,211 live more than one mile from any fixed naloxone access point, 902,611 can neither walk to supply nor readily reach it by transit, and exactly one fixed 24-hour access point serves the entire county. These figures involve no modeling, no smoothing, and no interpolation. They are counts of people and computations of distance, and they are conservative, because they credit as supply a set of locations whose walk-in public access is materially limited.

That the deficit is measurable matters because so little else about this crisis is measurable at fine geography here. Texas confidentiality rules suppress overdose death counts in 592 of 645 census tracts and in 138 of 150 federal reporting units, and no public state ZIP-level mortality product exists at all. A prioritization framework built primarily on that mortality signal would be building on estimates layered over suppressed counts. Inverting the approach, treating access as the measured quantity and mortality as one modeled input among five, produces recommendations that survive the data environment rather than depending on it. The strongest

evidence for that claim is internal: the placement ranking derived without any mortality input agrees with the ranking that includes it at Spearman rho = 0.966, and shares 45 of its top 50 sites.

The deficit is also tractable at the margin. Twenty-five well-chosen everyday storefronts, ordinary neighborhood retail concentrated in two northwest corridors and the southern and western Tier 1 geographies, would place supply within a half-mile walk of 231,806 residents who currently have none. That intervention is lawful today under the Texas standing order framework and requires no legislative change.

The barriers that local action cannot resolve remain what they were. Texas is one of three states without comprehensive Good Samaritan protection, fentanyl test strips remain criminalized after unanimous House passage, no syringe services program is authorized in the county, Medicaid remains unexpanded, and more than 90% of the county's harm reduction budget rests on federal funding now under proposed reduction. The comparative evidence indicates these are not incidental features of the county's overdose trajectory.

Two limitations bear final emphasis. No original primary data were collected, and the cultural and implementation analyses rest on published literature and existing community assessments rather than author-conducted fieldwork; future work should integrate participatory methods with community members in the highest-priority geographies. And the suppression regime that shaped this study's design also bounds its conclusions: the finest honest mortality geography available for Dallas County remains 150 aggregation units, 92% of them with suppressed counts. Improving that resolution would require access to case-level medical examiner records currently available only by request. Until then, the distance between what can be counted about supply and what can only be estimated about death is itself among the more consequential findings this analysis reports.

## References

Agency for Toxic Substances and Disease Registry. (2022). *CDC/ATSDR Social Vulnerability Index: Documentation and data*. Centers for Disease Control and Prevention. https://www.atsdr.cdc.gov/placeandhealth/svi/documentation/SVI_documentation_2020.html

Al-Awadhi, A. A., & Vegiraju, S. M. (2026). Geographic disparities in the opioid overdose crisis: Access, treatment, and mortality. *Cureus, 18*(1), e102184. https://doi.org/10.7759/cureus.102184

Baker Institute Drug Policy Program. (2025). *Accountability and transparency in Texas opioid settlement spending*. Rice University. https://www.bakerinstitute.org/research/accountability-and-transparency-texas-opioid-settlement-spending

Baumgartner, M., & Ledat, C. (2024). *Overdose Data to Action: Dallas County community needs assessment*. Dallas County Health and Human Services & Recovery Resource Council. https://www.dallascounty.org/Assets/uploads/docs/hhs/public-health/alcohol-substance/OD2A-2024-Community-Needs-Assessment.pdf

Be Well Texas. (2024, June 18). *Texas revamps Narcan distribution following delays, unpredictable supply*. https://bewelltexas.org/texas-revamps-narcan-distribution-following-delays-unpredictable-supply/

Brookings Institution. (2025, July 21). *Progress under threat: The future of overdose prevention in the United States*.

https://www.brookings.edu/articles/progress-under-threat-the-future-of-overdose-prevention-in-the-united-states/

Centers for Disease Control and Prevention. (2024). *Strengthening syringe services programs*. https://www.cdc.gov/hepatitis-syringe-services/php/about/index.html

Centers for Disease Control and Prevention. (2025). *A lifeline for those in need: Dallas County's overdose prevention hotline*. https://www.cdc.gov/overdose-prevention/php/od2a/dallas.html

Centers for Disease Control and Prevention, National Center for Health Statistics. (2026a). *VSRR provisional drug overdose death counts* [Data set]. https://data.cdc.gov/d/gb4e-yj24

Centers for Disease Control and Prevention, National Center for Health Statistics. (2026b). *Model-based drug overdose death rates by census tract* [Data set]. https://data.cdc.gov/d/4day-mt2f

Claborn, K., Samora, J., McCormick, K., Whittfield, Q., Courtois, F., Lozada, K., Sledge, D., Burwell, A., Chavez, S., Bailey, J., Bailey, C., Dalton Pederson, C., Zagorski, C., Hill, L., Conway, F. N., Holleran Steiker, L., Cance, J., & Potter, J. (2023). "We do it ourselves": Strengths and opportunities for improving the practice of harm reduction. *Harm Reduction Journal, 20*(1), 70. https://doi.org/10.1186/s12954-023-00809-7

D Magazine. (2021, November 15). *The cost of 'pill culture' and the opioid epidemic in Dallas County*. https://www.dmagazine.com/frontburner/2021/11/the-cost-of-pill-culture-and-the-opioid-epidemic-in-dallas-county/

Dahlem, C. H. G., English, E., Dwan, M., Carlson, C., Dora-Laskey, A., Kocher, K. E., & Losman, E. D. (2025). Implementation of a naloxone vending machine in the public

space of an emergency department: A feasibility and acceptability study. *Journal of Emergency Nursing, 51*(4), 607–615. https://doi.org/10.1016/j.jen.2025.03.009

Dallas Area Rapid Transit. (2026). *General Transit Feed Specification* [Data set]. https://www.dart.org/about/about-dart/developer-resources

Dallas County Health and Human Services. (n.d.-a). *Alcohol and substance use prevention: Harm reduction resources*. https://www.dallascounty.org/departments/dchhs/public-health/alcohol-substance/harm-reduction-resources.php

Dallas County Health and Human Services. (n.d.-b). *STI/HIV*. https://www.dallascounty.org/departments/dchhs/public-health/sti-hiv/

Dallas County Health and Human Services. (2024). *Overdose Data to Action: 2024 annual surveillance report of preliminary trends in drug overdoses in Dallas County*. https://www.dallascounty.org/Assets/uploads/docs/hhs/public-health/alcohol-substance/reports/OD2A%20-%202024%20Annual%20Surveillance%20Report%20of%20Preliminary%20Trends%20in%20Drug%20Overdoses%20in%20Dallas%20County.pdf

Dallas Express. (2023, June 27). *Dallas joins drug overdose mapping project in effort to combat worsening opioid crisis*. https://dallasexpress.com/city/dallas-to-map-drug-overdoses/

Dallas Express. (2024, August 19). *Dallas opioid crisis worsens as deaths continue to climb*. https://dallasexpress.com/city/dallas-opioid-crisis-is-worsening-and-rehabilitation-resources-remain-in-short-supply/

de la Flor, C., Porsa, E., & Nijhawan, A. E. (2017). Opt-out HIV and hepatitis C testing at the Dallas County Jail: Uptake, prevalence, and demographic characteristics of testers. *Public Health Reports, 132*(6), 617-621. https://doi.org/10.1177/0033354917732755

Episcopal Health Foundation. (n.d.). *New evidence shows additional benefits of Medicaid expansion in Texas*. https://www.episcopalhealth.org/enews/new-evidence-shows-additional-benefits-medicaid-expansion-texas/

Ezell, J. M., Simek, E., Shetty, N., Pho, M. T., Bluthenthal, R. N., Goddard-Eckrich, D. A., & Choi, S. (2024). A scoping review of the utilization of opioid use treatment, harm reduction, and culturally tailored interventions among racial/ethnic minorities in the United States. *International Journal of Mental Health and Addiction*. https://doi.org/10.1007/s11469-024-01373-2

Febres-Cordero, S., Shasanmi-Ellis, R. O., & Sherman, A. D. F. (2023). Labeled as "drug-seeking": Nurses use harm reduction philosophy to reflect on mending mutual distrust between healthcare workers and people who use drugs. *Frontiers in Public Health, 11*, 1277562. https://doi.org/10.3389/fpubh.2023.1277562

Federal Reserve Bank of Dallas. (2025). *Has the opioid crisis peaked in Texas and the U.S.?* https://www.dallasfed.org/research/swe/2025/swe2516

Filter Magazine. (2025, April 15). *Texas bill could expand sterile syringe access — for a fee*. https://filtermag.org/texas-bill-syringe-service-programs/

Fischer, L. S., Asher, A., Stein, R., Becasen, J., Doreson, A., Mermin, J., Meltzer, M. I., & Edlin, B. R. (2025). Effectiveness of naloxone distribution in community settings to reduce opioid overdose deaths among people who use drugs: A systematic review and meta-analysis. *BMC Public Health, 25*, 1135. https://doi.org/10.1186/s12889-025-22210-8

Fomby, T. B. (n.d.). *DIVERT Court of Dallas County cost-benefit analysis*. Southern Methodist University. https://socialvalueuk.org/wp-content/uploads/2023/05/Divert-Court-of-Dallas-County_Cost-Benefit-Analysis-Texas.pdf

Hill, L. G., Evoy, K. E., Reveles, K. R., & Martin, B. C. (2019). State-level approaches to expanding pharmacists' authority to dispense naloxone may affect accessibility. *JAMA Internal Medicine, 179*(10), 1442–1443. https://doi.org/10.1001/jamainternmed.2019.2978

Housing Forward North Texas. (2026). *2026 point-in-time count: Dallas and Collin counties*. https://www.fightinghomelessness.org

Houston Landing. (2025, May 18). *Amid hundreds of deaths each year, here's how Houston is spending opioid settlement dollars*. https://www.houstonlanding.org/amid-hundreds-of-deaths-each-year-heres-how-houston-is-spending-opioid-settlement-dollars/

Houston Public Media. (2024, December 23). *Report: Harris County has significantly increased substance-involved deaths and healthcare visits*. https://www.houstonpublicmedia.org/articles/news/health-science/2024/12/23/509463/report-harris-county-has-significantly-increased-substance-involved-deaths-and-healthcare-visits/

Ilyas, A. I., & Ilyas, S. (2026). *dallas-naloxone-access: Measured block-level naloxone access and placement analysis for Dallas County, Texas* (Version 1.0.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.21537399

Jenks, G. F. (1967). The data model concept in statistical mapping. *International Yearbook of Cartography, 7*, 186–190.

Kaiser Family Foundation. (2025). *Implications of potential federal Medicaid reductions for addressing the opioid epidemic*. https://www.kff.org/medicaid/implications-of-potential-federal-medicaid-reductions-for-addressing-the-opioid-epidemic/

KVUE News. (2025, April 22). *Texas House unanimously gives initial approval to bill that would legalize fentanyl testing strips*. https://www.kvue.com/article/news/politics/texas-legislature/fentanyl-testing-strip-legalization-texas-house/269-ad881c3d-75d1-4e8d-a211-eb26f3a7c55a

Law Office of David D. White. (2024). *New drug laws in Texas 2024*. https://www.wm-attorneys.com/new-drug-laws-texas-2024/

Liang, D., Shi, Y., & Rice, T. (2023). Association of medication-assisted therapy and risk of drug overdose-related hospitalization or emergency room visits in patients with opioid use disorder. *Drug and Alcohol Dependence, 251*, 110944. https://doi.org/10.1016/j.drugalcdep.2023.110944

McGinty, E. E., & Barry, C. L. (2020). Stigma reduction to combat the addiction crisis: Developing an evidence base. *New England Journal of Medicine, 382*(14), 1291–1292. https://doi.org/10.1056/NEJMp2000227

Medscape. (2026, February 11). *Retention rates remain low for US opioid treatment programs*. https://www.medscape.com/viewarticle/retention-rates-remain-low-us-opioid-treatment-programs-2026a10004bs

NAMA Recovery. (2024). *Federal regulations on methadone treatment*. https://www.namaweb.org

NASEN Directory. (2024). *Dallas Harm Reduction Aid*. https://nasen.org/sep/dallas-harm-reduction-aid

National Academy for State Health Policy. (2025). *State opioid settlement spending decisions: Texas*. https://nashp.org/state-tracker/state-opioid-settlement-spending-decisions/texas/

NEXT Distro. (n.d.). *Free Narcan and naloxone in Philadelphia*. https://nextdistro.org/philly

Philadelphia Department of Public Health. (2025, April 29). *Health department releases reports on overdose crisis*. https://www.phila.gov/2025-04-29-health-department-releases-reports-on-overdose-crisis/

Philadelphia Department of Public Health. (2026, May 11). *Unintentional drug overdose fatalities in Philadelphia in 2024*. https://www.phila.gov/2026-05-11-unintentional-drug-overdose-fatalities-in-philadelphia-in-2024/

Philadelphia Inquirer. (2024, September 12). *Philly poverty rate sees largest drop in 10 years, but we're still the poorest big city*. https://www.inquirer.com/politics/philadelphia/philadelphia-poverty-rate-decline-household-income-20240912.html

Philadelphia Inquirer. (2025, December 31). *Overdose deaths in Philly on track to be the lowest in nearly 10 years*. https://www.inquirer.com/health/overdose-deaths-philadelphia-pennsylvania-2025-20251230.html

Ruhm, C. J. (2018). Corrected US opioid-involved drug poisoning deaths and mortality rates, 1999–2015. *American Journal of Public Health, 108*(12), 1626–1628. https://doi.org/10.2105/AJPH.2018.304592

Substance Abuse and Mental Health Services Administration. (2026). *Findtreatment.gov treatment facility locator* [Data set]. https://findtreatment.gov

Substance Use Philly. (n.d.). *Get free naloxone and fentanyl test strips in Philly*. https://www.substanceusephilly.com/harm-reduction

Texas Alcoholic Beverage Commission. (2026). *TABC licenses and permits* [Data set]. https://data.texas.gov/d/kguh-7q9z

Texas Community Health News. (2020). *Dallas County*. https://www.texascommunityhealthnews.org/counties/dallas/

Texas Comptroller of Public Accounts. (n.d.). *About the opioid settlements*. https://comptroller.texas.gov/programs/opioid-council/settlements/index.php

Texas Criminal Justice Exchange. (n.d.). *Harm reduction facts*. https://texascje.org/harm-reduction-facts

Texas Department of State Health Services. (2026). *Texas Overdose Data to Action: Naloxone distribution interactive map*. https://healthdata.dshs.texas.gov/dashboard/drugs-and-alcohol/all-drugs/naloxone-distribution-interactive-map

Texas Health and Human Services. (n.d.). *Texas Targeted Opioid Response*. https://www.hhs.texas.gov/services/mental-health-substance-use/adult-substance-use/texas-targeted-opioid-response

Texas Health and Safety Code § 481.002. *Definitions*. Texas Legislature Online. https://statutes.capitol.texas.gov/Docs/HS/htm/HS.481.htm

Texas Health and Safety Code § 481.115. *Offense: Possession of substance in Penalty Group 1*. Texas Legislature Online. https://statutes.capitol.texas.gov/Docs/HS/htm/HS.481.htm

Texas Health and Safety Code § 481.115(g)-(i) (2021). *Jessica Sosa Act: Defense to prosecution for persons seeking emergency assistance during a drug overdose*. Texas Legislature Online. https://statutes.capitol.texas.gov/Docs/HS/htm/HS.481.htm

Texas Health and Safety Code § 481.1151 (2023). *Penalty Group 1-B: Fentanyl and fentanyl analogs*. Texas Legislature Online. https://statutes.capitol.texas.gov/Docs/HS/htm/HS.481.htm

Texas Legislature. (2025). *House Bill 1644: Relating to the possession and use of drug test equipment*. Texas Legislature Online. https://capitol.texas.gov/BillLookup/History.aspx?LegSess=89R&Bill=HB1644

Texas Monthly. (2024, July 25). *Texas refuses to expand Medicaid. Here's what we're missing out on*. https://www.texasmonthly.com/news-politics/medicaid-expansion-benefits-legislature/

Texas Observer. (2025, May 6). *Will Dan Patrick's Senate stymie fentanyl test strip legalization (again)?* https://www.texasobserver.org/fentanyl-test-strips-texas-legislature/

Texas Overdose Training Initiative. (2025). *Naloxone standing order FAQ*. https://www.txoti.org/assets/NSO_FAQ.pdf

Texas Senate. (2023). *Senate Bill 1319: Relating to the reporting of certain overdose information and the mapping of overdoses for public safety purposes*. https://senate.texas.gov/news.php?id=20230315a

Tough, P. (2025, October 7). *$68 million COVID-era federal funding loss and other potential cuts threaten Texas opioid fight*. Texas Tribune. https://www.texastribune.org/2025/10/07/texas-naloxone-narcan-distribution-federal-budget-cuts/

U.S. Census Bureau. (2021). *2020 Census redistricting data (P.L. 94-171) summary file* [Data set]. https://www.census.gov/programs-surveys/decennial-census/about/rdo/summary-files.html

U.S. Census Bureau. (2024). *TIGER/Line shapefiles* [Data set]. https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html

U.S. Census Bureau. (2026). *American Community Survey 2020–2024 5-year estimates* [Data set]. https://www.census.gov/programs-surveys/acs

U.S. Government Accountability Office. (2021). *Drug misuse: Most states have Good Samaritan laws and research indicates they may have positive effects* (GAO-21-248). https://www.gao.gov/products/gao-21-248

Vadiei, N., Axon, D. R., & Eckert, B. (2025). Naloxone distribution models in the United States: A scoping review. *Substance Use & Addiction Journal*. https://doi.org/10.1177/29767342241289008

Walley, A. Y., Xuan, Z., Hackman, H. H., Quinn, E., Doe-Simkins, M., Sorensen-Alawad, A., Ruiz, S., & Ozonoff, A. (2013). Opioid overdose rates and implementation of overdose education and nasal naloxone distribution in Massachusetts: Interrupted time series analysis. *BMJ, 346*, f174. https://doi.org/10.1136/bmj.f174

Ware, O. D., Garcia-Romeu, A., Zamarripa, C. A., Hughes, T., Wager, L., & Spindle, T. (2024). Codeine and promethazine: Exploratory study on "lean" or "sizzurp" using national survey data and an online forum. *PLOS ONE, 19*(3), e0301024. https://doi.org/10.1371/journal.pone.0301024

WHYY. (2024, October 2). *Drug overdose deaths among Philadelphians fall for first time in 5 years*. https://whyy.org/articles/philadelphia-overdose-deaths-decline/

World Population Review. (2026). *Philadelphia, Pennsylvania population 2026*. https://worldpopulationreview.com/us-cities/pennsylvania/philadelphia